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TATTATT

The Human Genome



2003

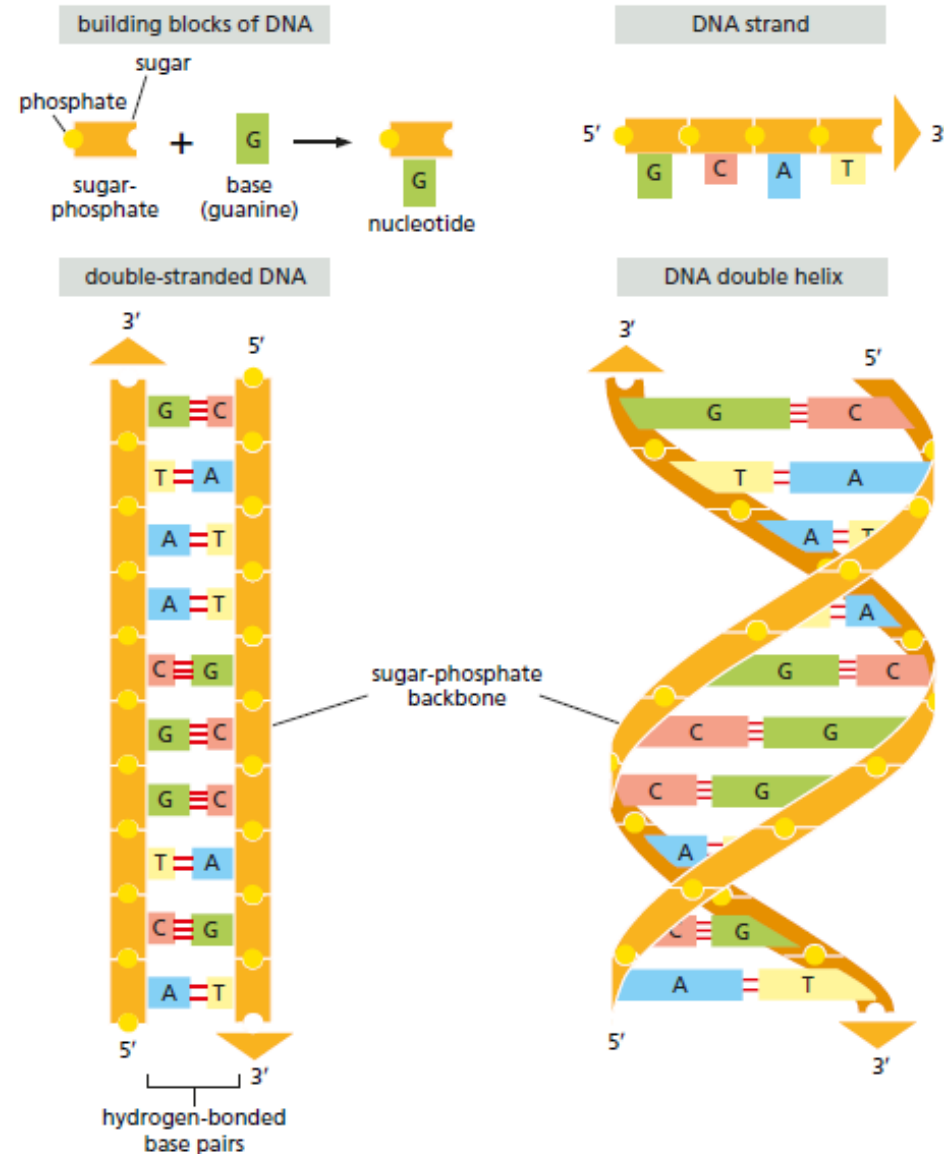
“Fifteen years, six countries, twenty centers. Three billion dollars, three billion letters. One dollar per letter – such a deal!”

Eric Lander's seven-word “ Nano Lecture”

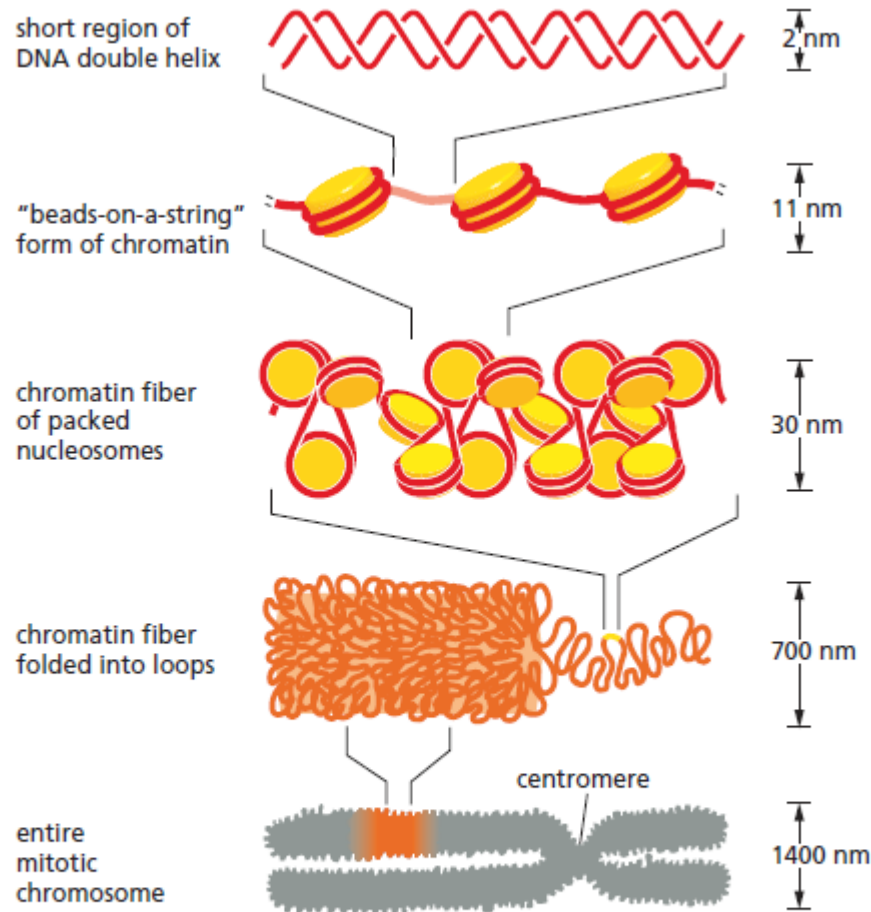
**“Genome: bought the book;
hard to read”**

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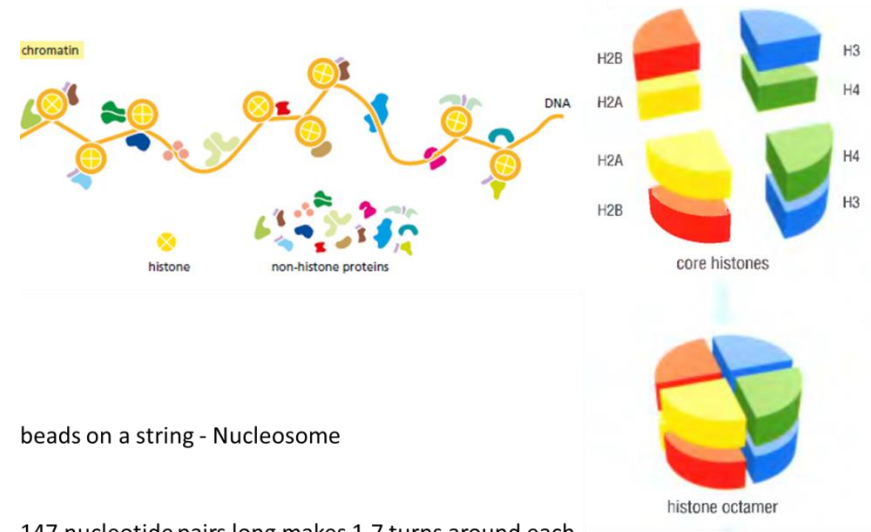
DNA and its building blocks



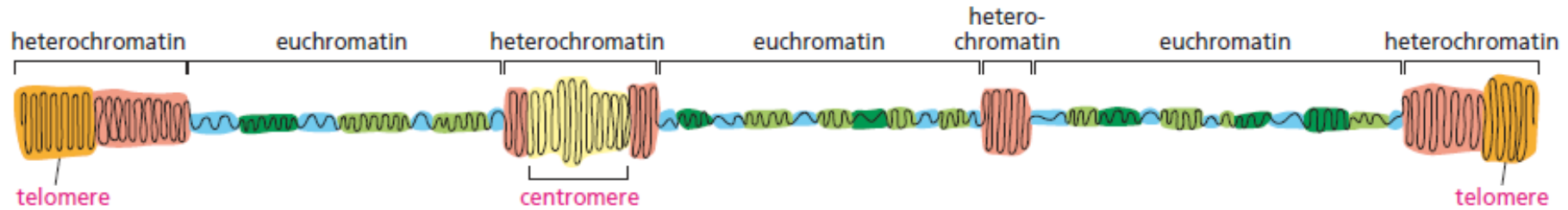
The organization of Chromosomes



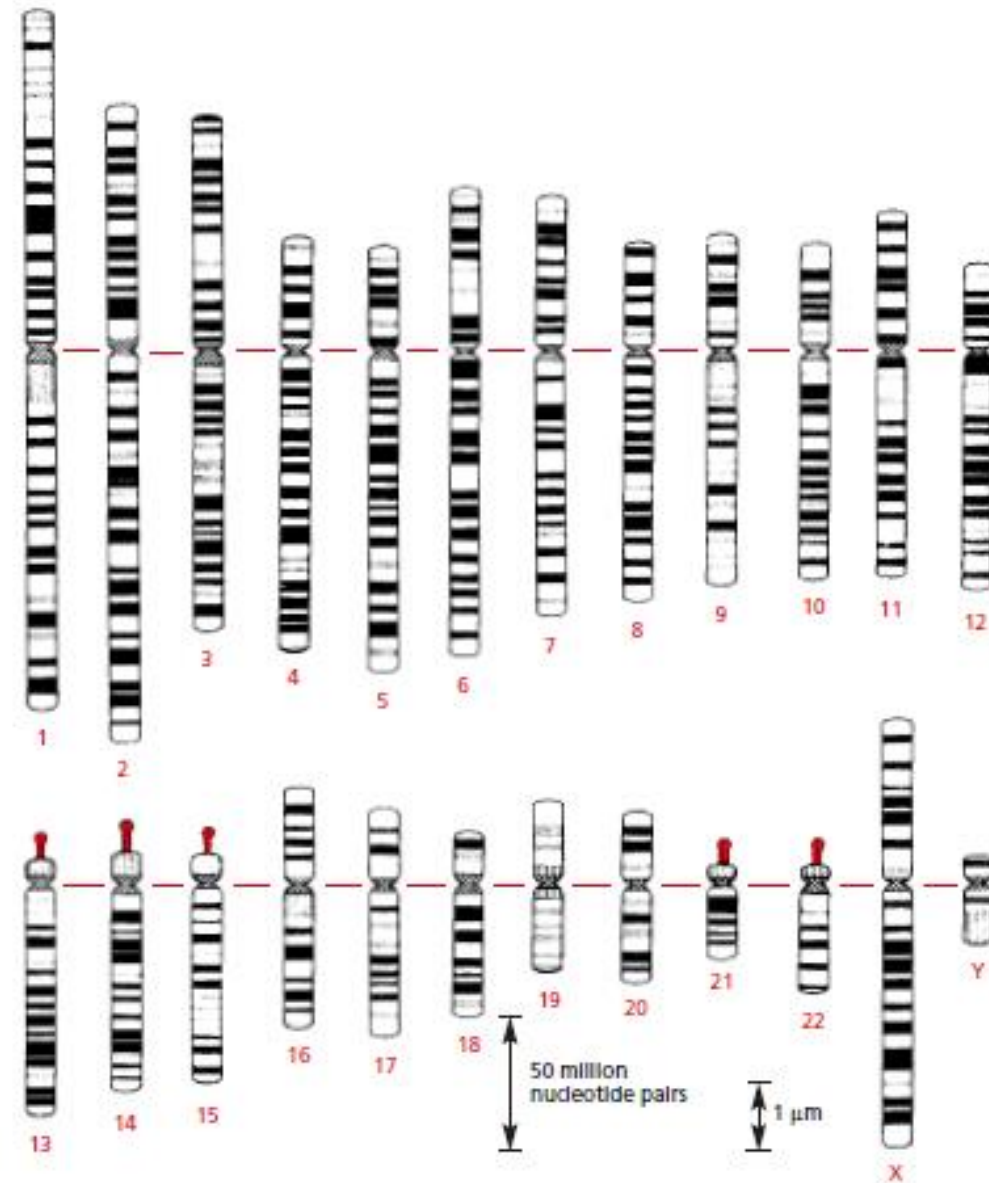
NET RESULT: EACH DNA MOLECULE HAS BEEN PACKAGED INTO A MITOTIC CHROMOSOME THAT IS 10,000-FOLD SHORTER THAN ITS FULLY EXTENDED LENGTH



The structure of chromatin varies



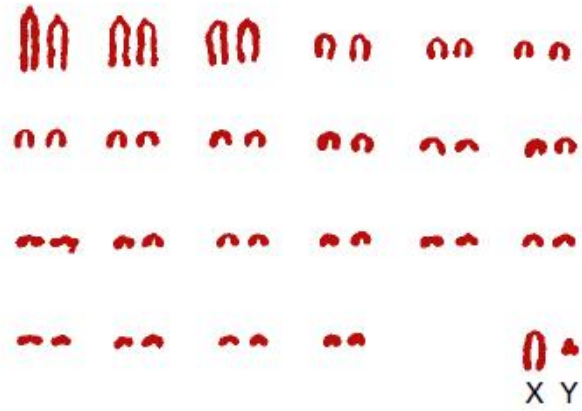
The banding patterns of human chromosomes



Giemsa Staining



Chinese muntjac



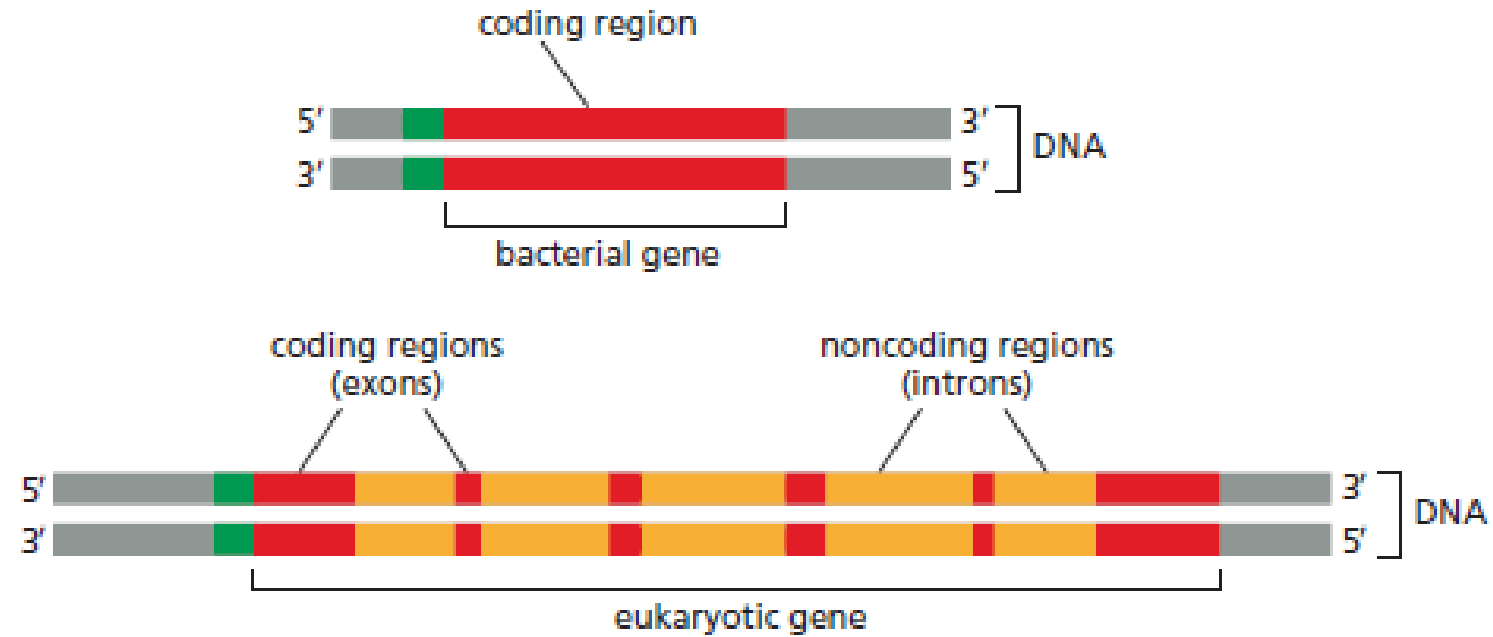
Indian muntjac



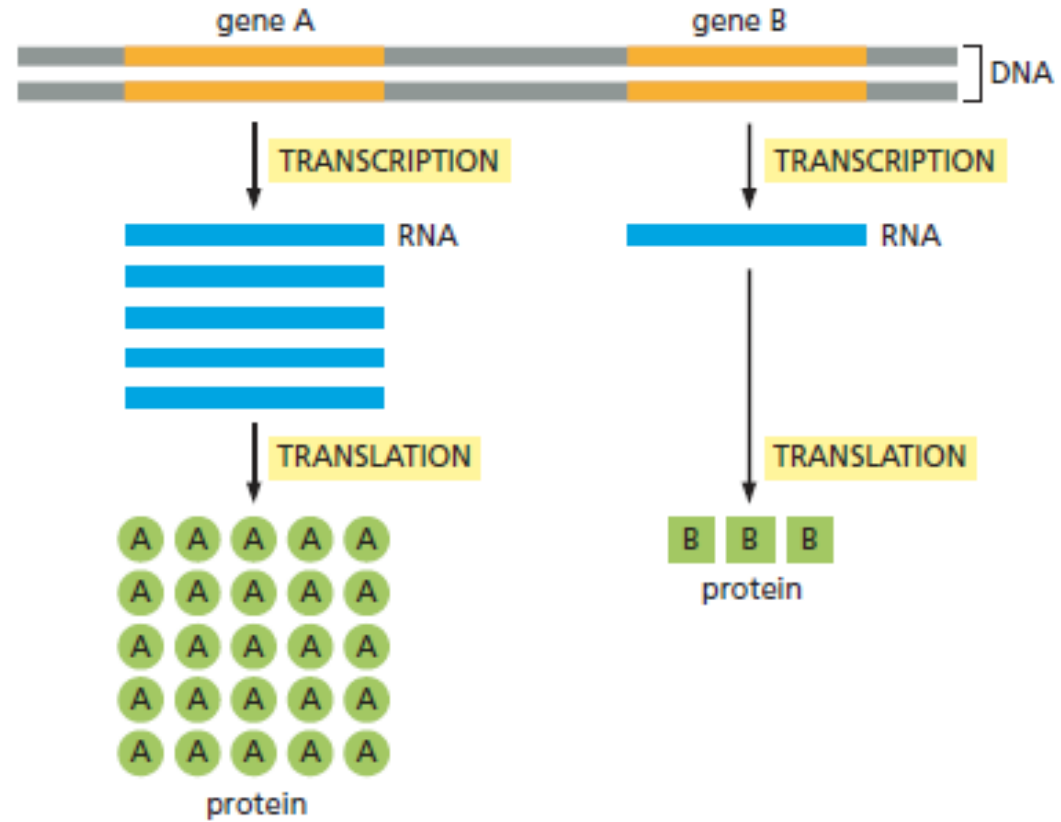
Interesting observations from Beatrice !

- ❑ In some insects like grasshoppers, crickets, and roaches, there is only one kind of sex chromosome, X. If you get 2 of them, you're female, and if you just get one, you're a male.
- ❑ In birds (and even some insects and fishes) it's the females that have different sex chromosomes. Females are ZW and males are ZZ. Scientists changed the letters to differentiate between the XX-female, XY-male scenario.
- ❑ In ants and bees, there are no sex chromosomes. Instead, sex is determined by whether or not an egg was fertilized. If the egg isn't fertilized, the offspring is male. If the egg is fertilized, it's female. So male ants have no fathers, and they have half as many chromosomes as females. Poor little things!
- ❑ A species of deer called the Indian muntjac (as opposed to the Chinese muntjac) has three different kinds of sex chromosomes—X and two versions of Y, called Y1 and Y2. Females are XX, and males are XY1Y2. Ha, those crazy muntjacs.
- ❑ May we never forget that biology is a complicated, messy, random process that has found several ways for dealing with sex and reproduction (and just about every other bit of life business). There is really no rhyme or reason for any of it. If it works, it works. If it doesn't, you die. That's pretty much the only rule in nature that applies all the time.

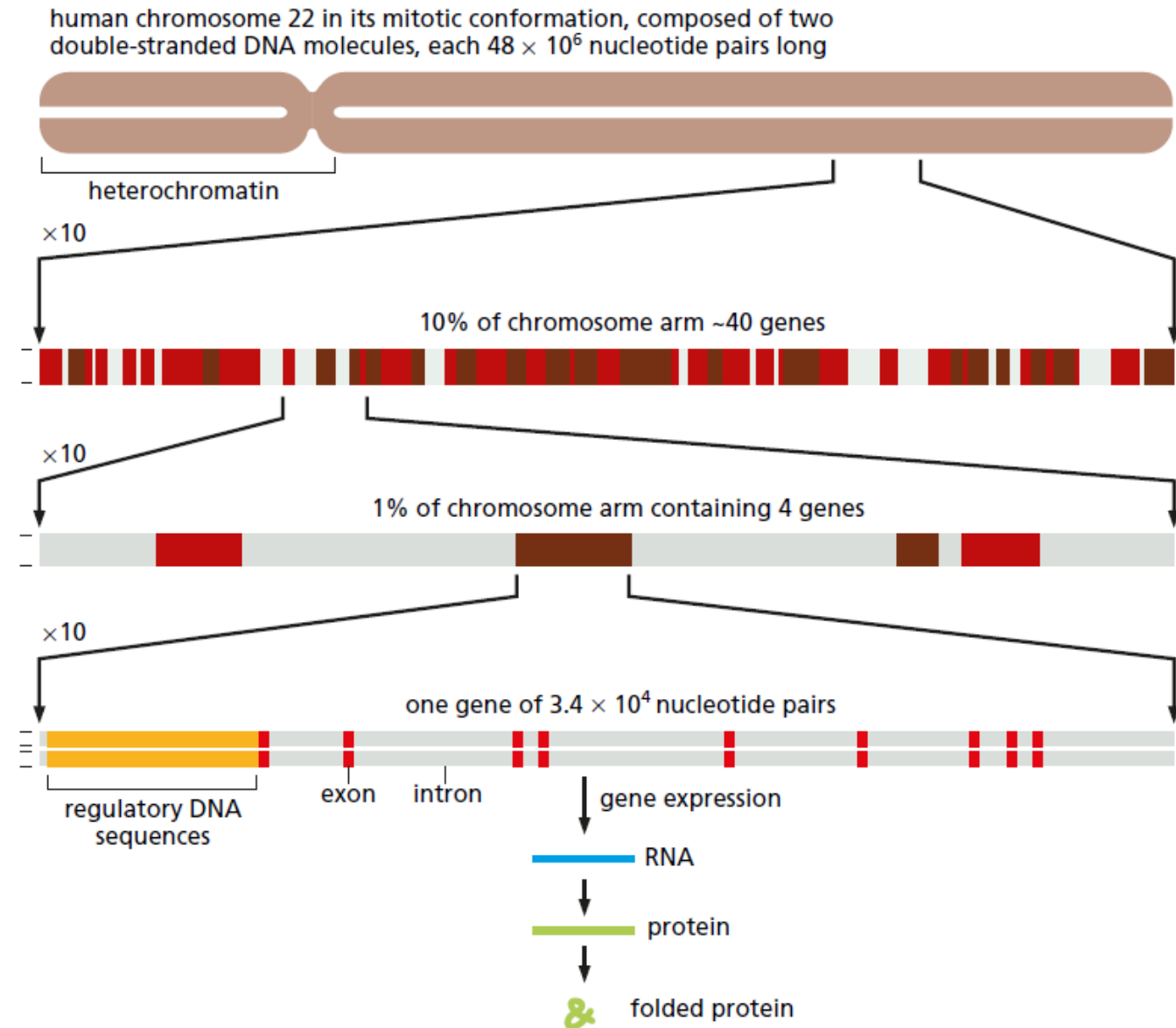
Eukaryotic and bacterial genes are organized differently



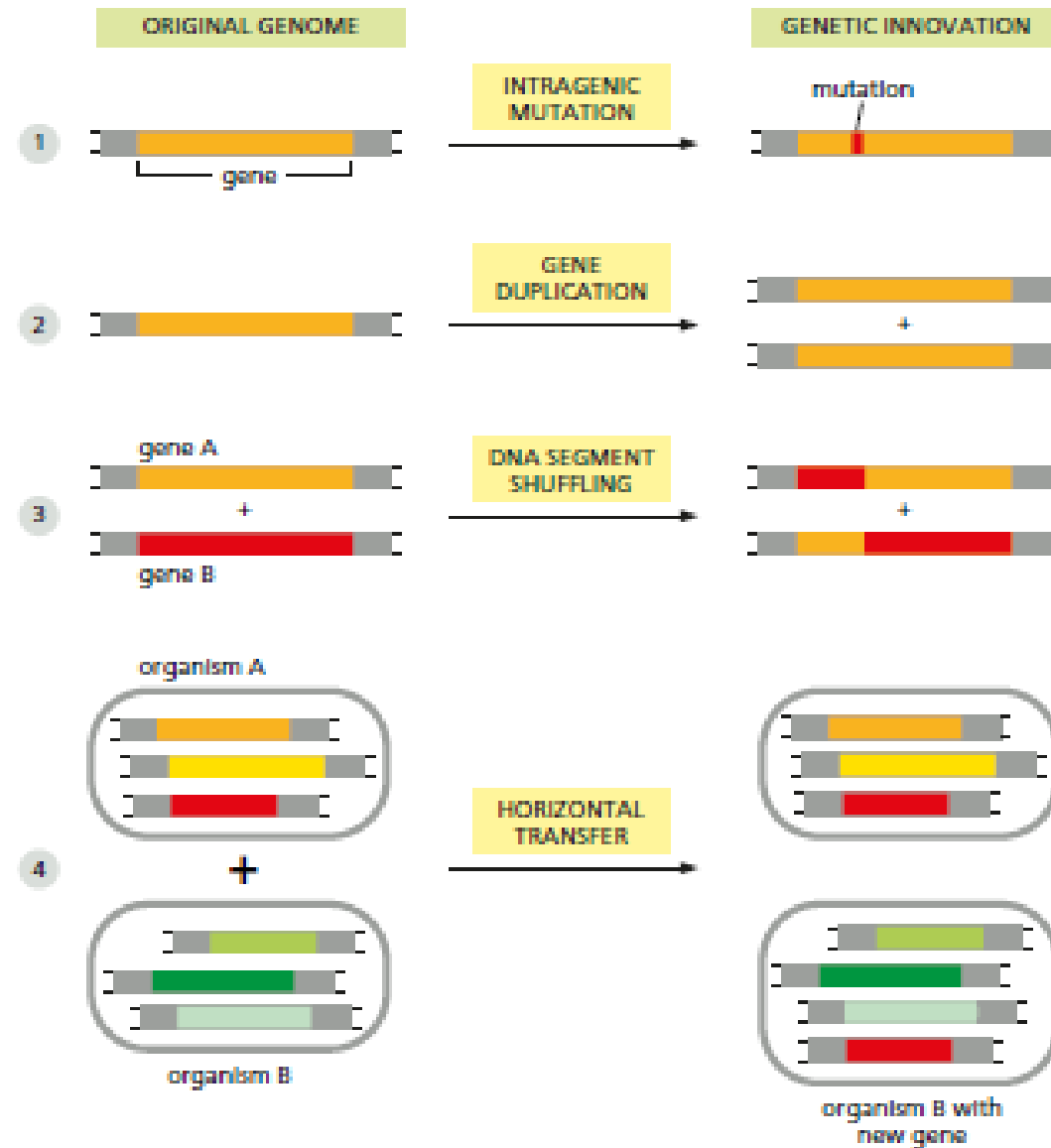
How Cells Read the Genome



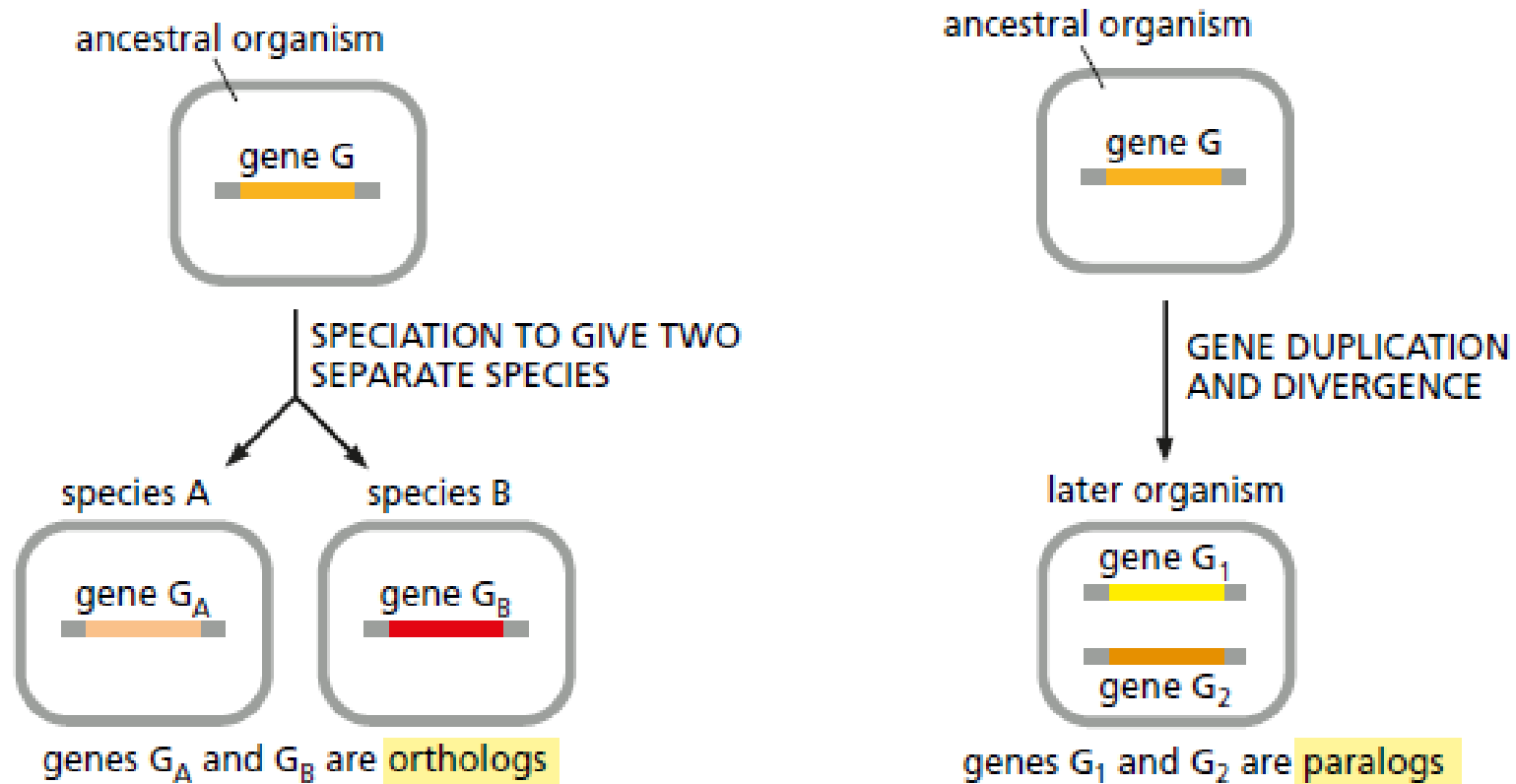
The organization of genes on a human chromosome



New genes are generated from existing genes

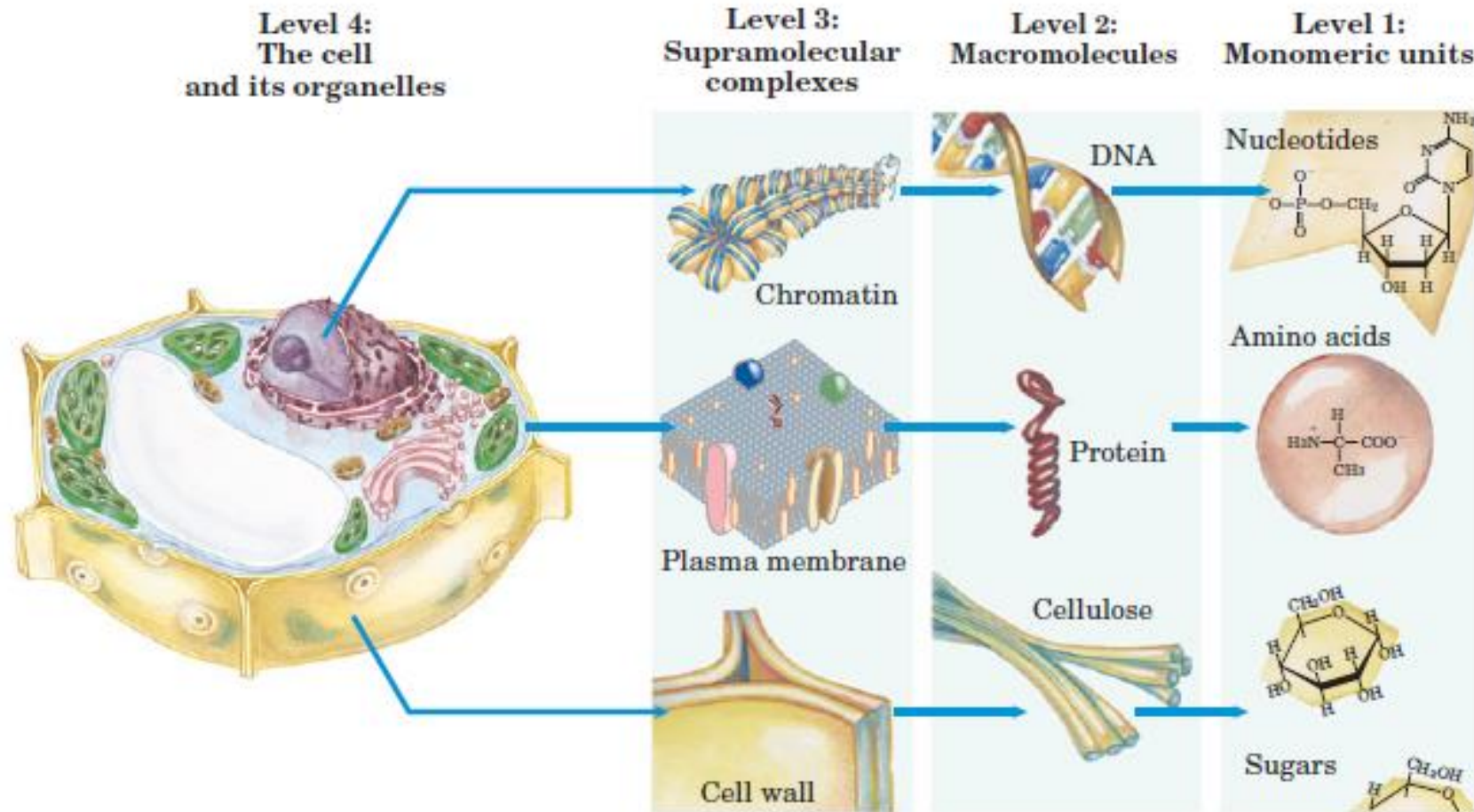


Gene duplications give rise to families of genes

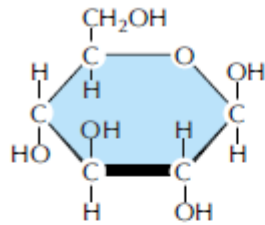




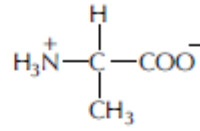
Structural hierarchy in the molecular organization of cells



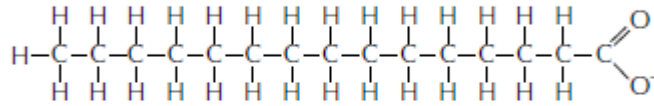
Cells function as biochemical factories



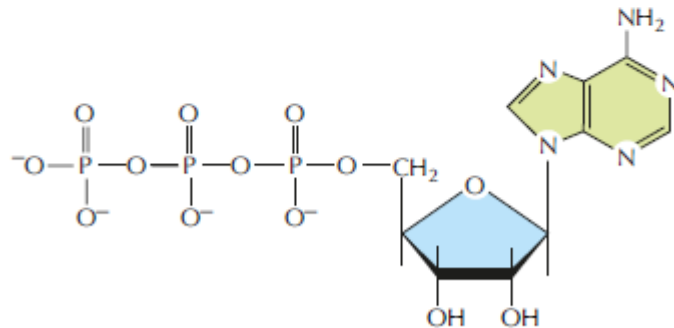
A SUGAR



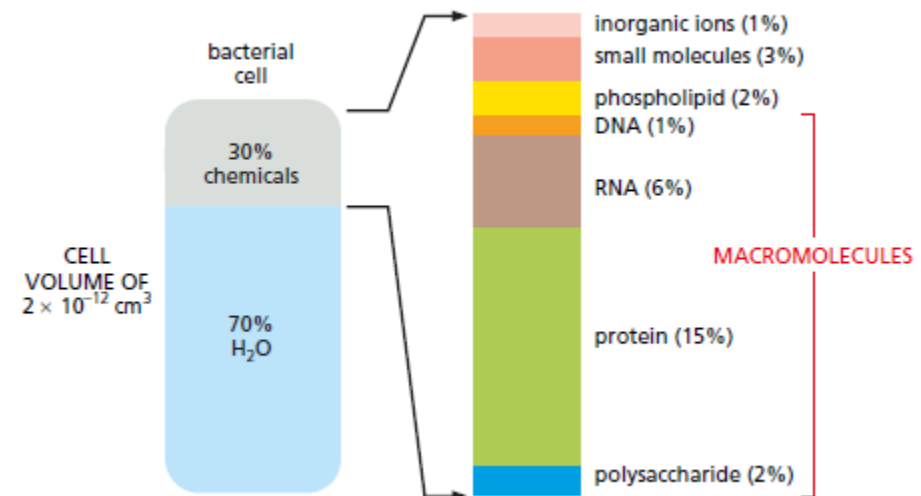
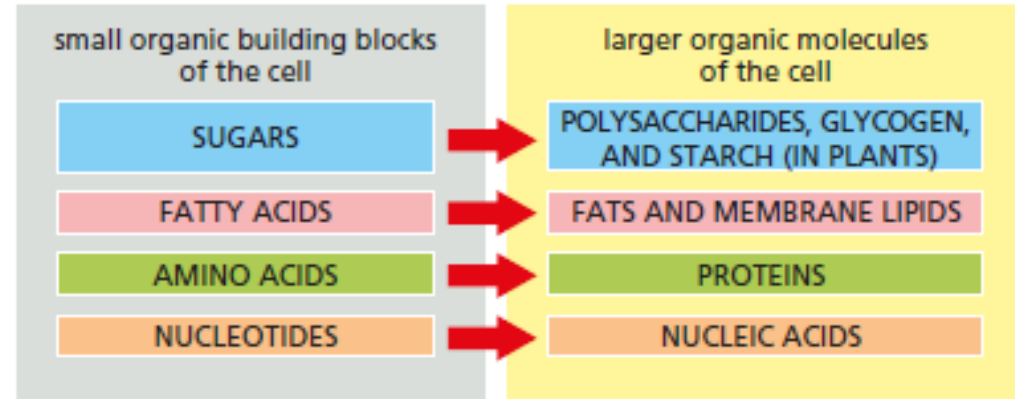
AN AMINO ACID



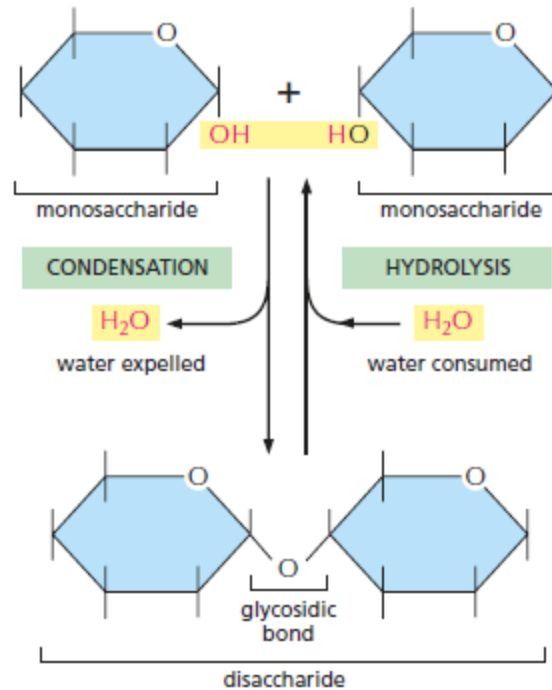
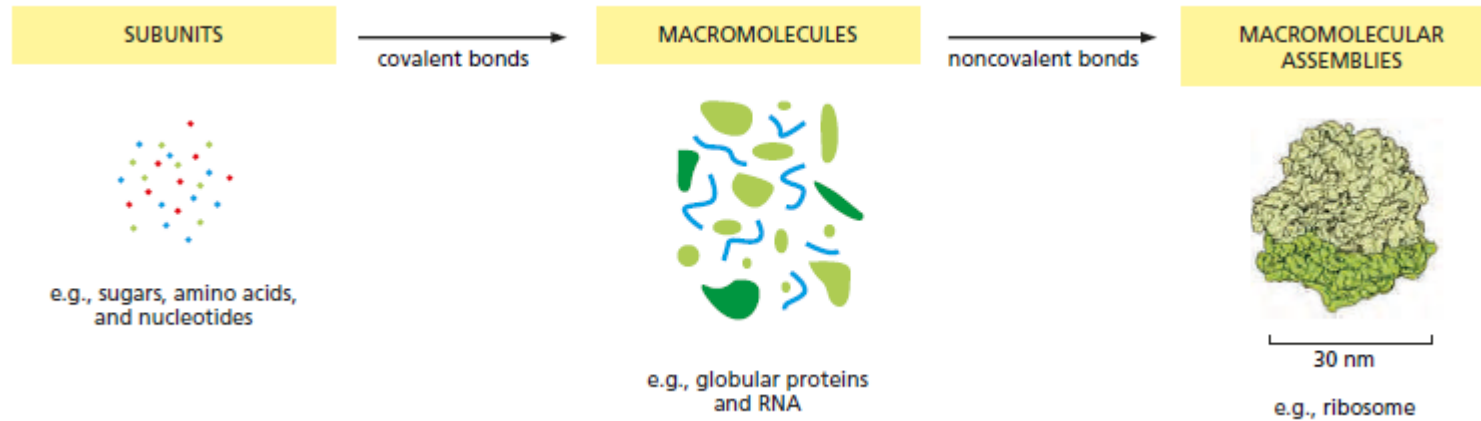
A FATTY ACID



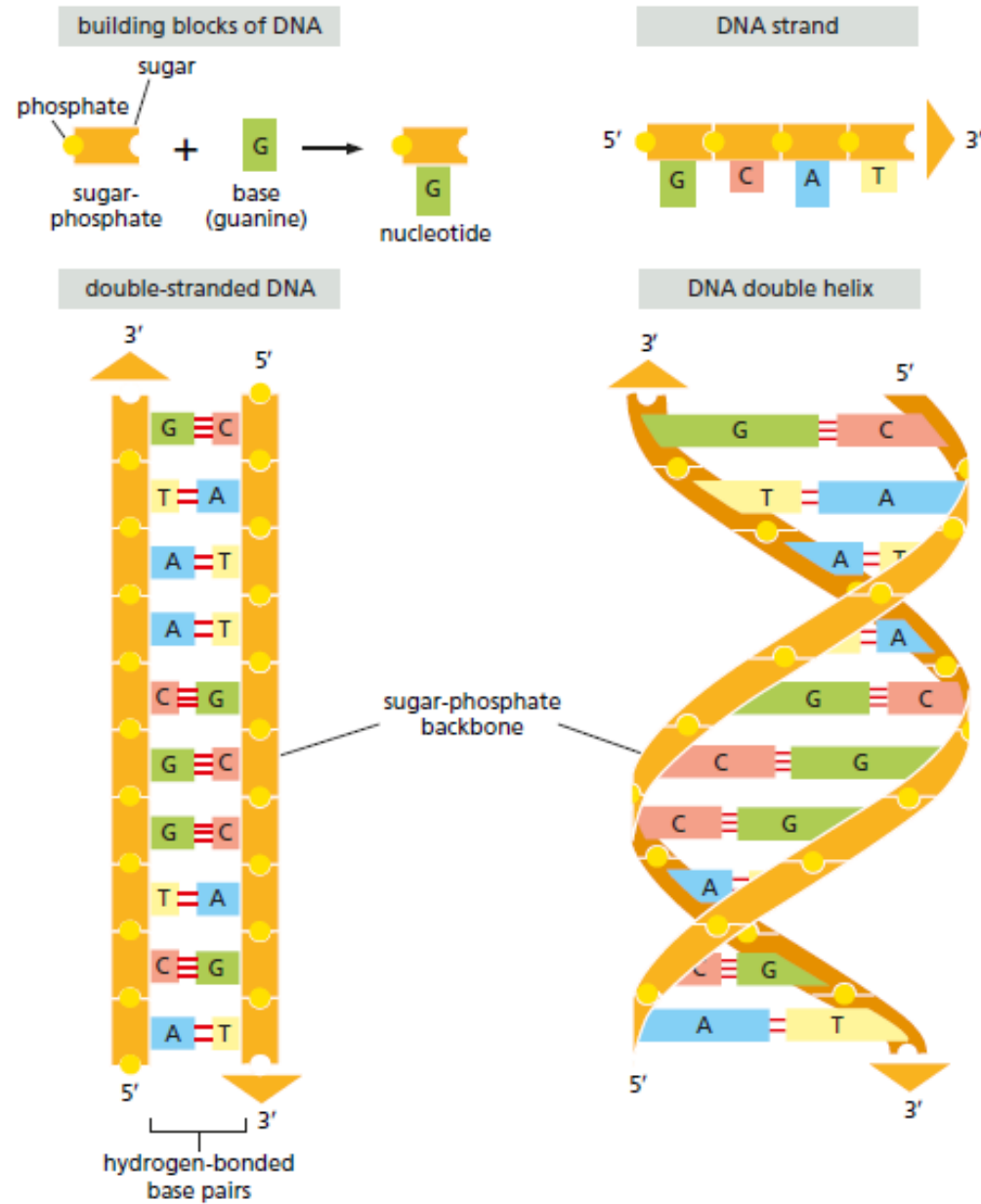
A NUCLEOTIDE



Biochemical bond formation

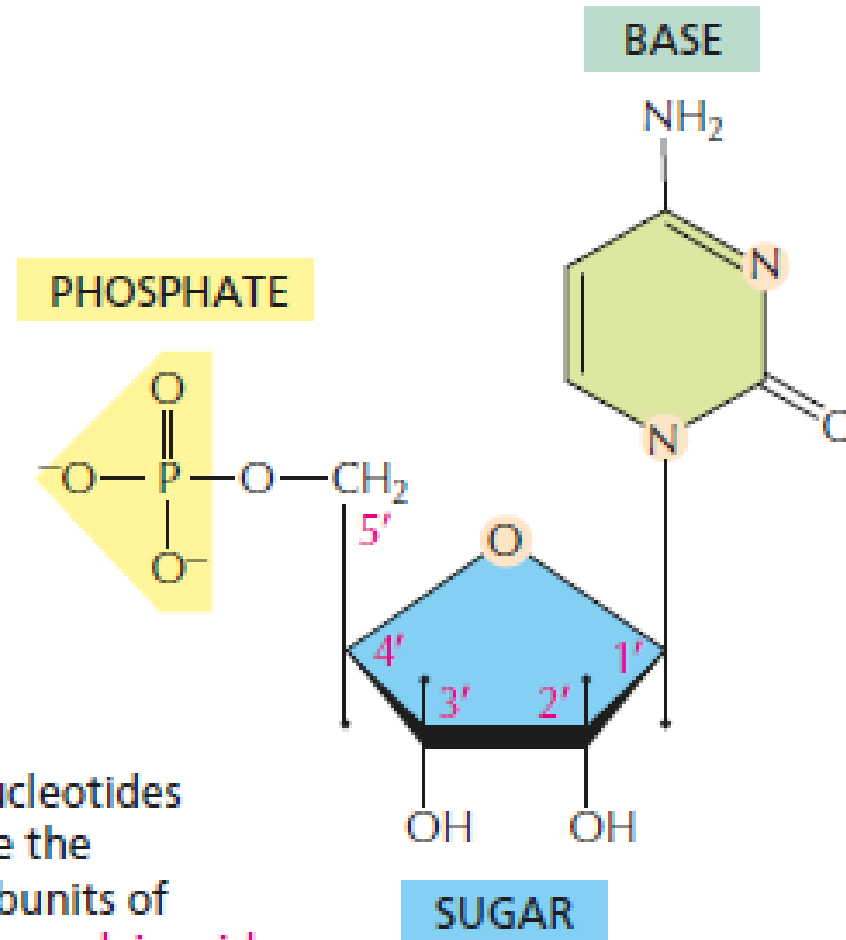


DNA and its building blocks



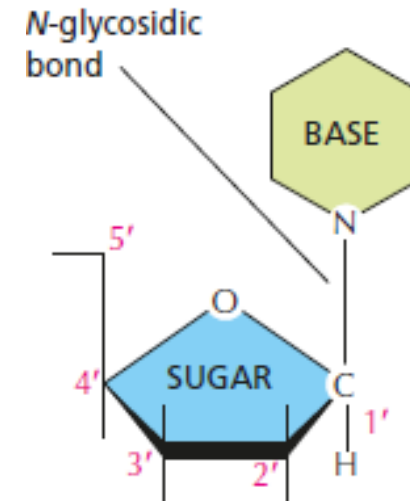
Nucleotides

A nucleotide consists of a nitrogen-containing base, a five-carbon sugar, and one or more phosphate groups.



Nucleotides are the subunits of the nucleic acids.

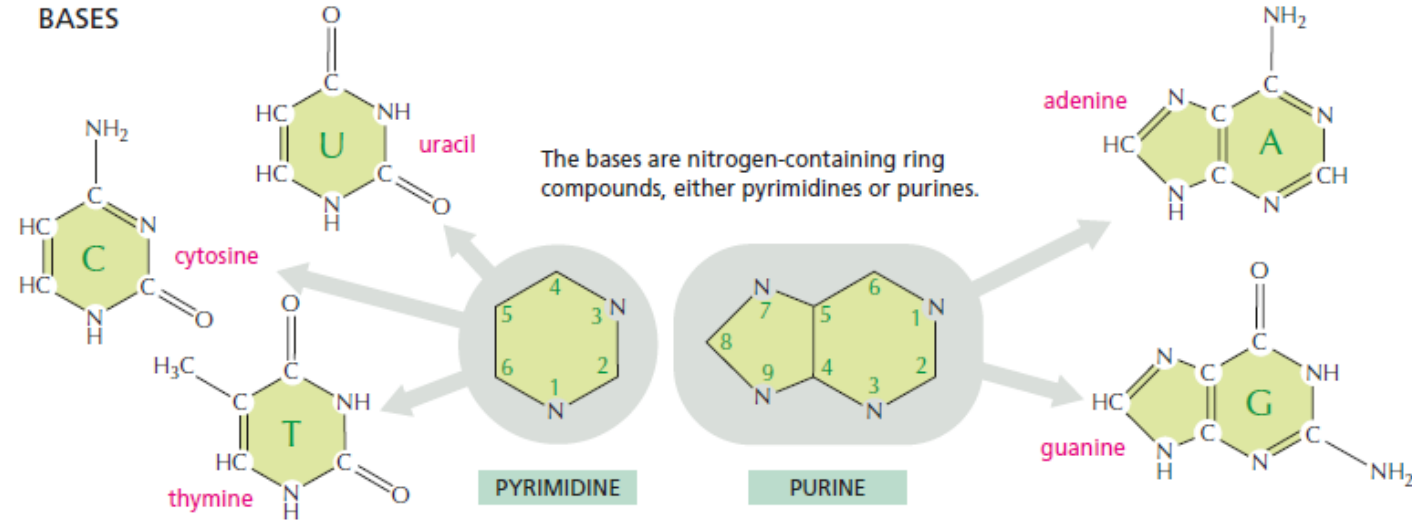
BASE-SUGAR LINKAGE



The base is linked to the same carbon (C1) used in sugar-sugar bonds.

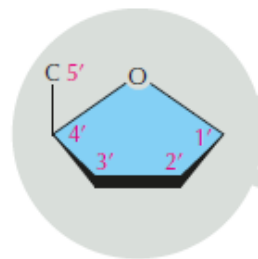
Nucleotides

BASES

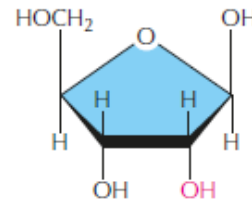


SUGARS

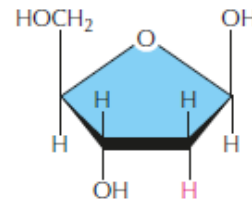
PENTOSE
a five-carbon sugar



two kinds of pentoses are used



β -D-ribose
used in ribonucleic acid (RNA)

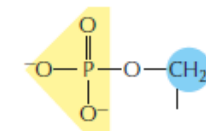


β -D-2-deoxyribose
used in deoxyribonucleic acid (DNA)

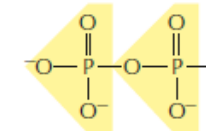
Each numbered carbon on the sugar of a nucleotide is followed by a prime mark; therefore, one speaks of the "5-prime carbon," etc.

PHOSPHATES

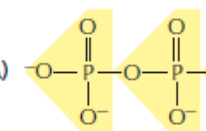
The phosphates are normally joined to the C5 hydroxyl of the ribose or deoxyribose sugar (designated 5'). Mono-, di-, and triphosphates are common.



as in
AMP



as in
ADP



as in
ATP

The phosphate makes a nucleotide negatively charged.

Nucleotides vs Nucleoside

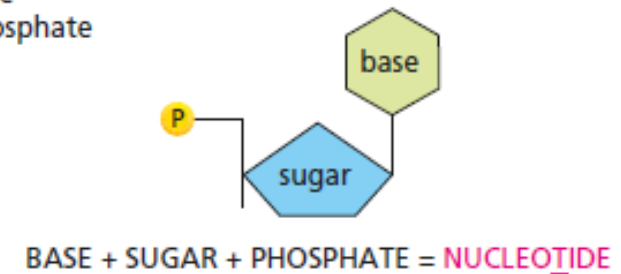
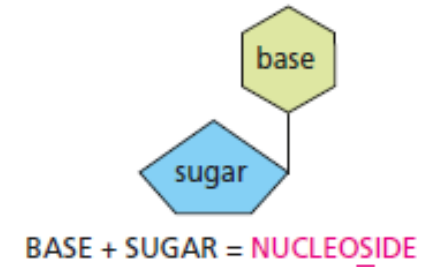
NOMENCLATURE

The names can be confusing, but the abbreviations are clear.

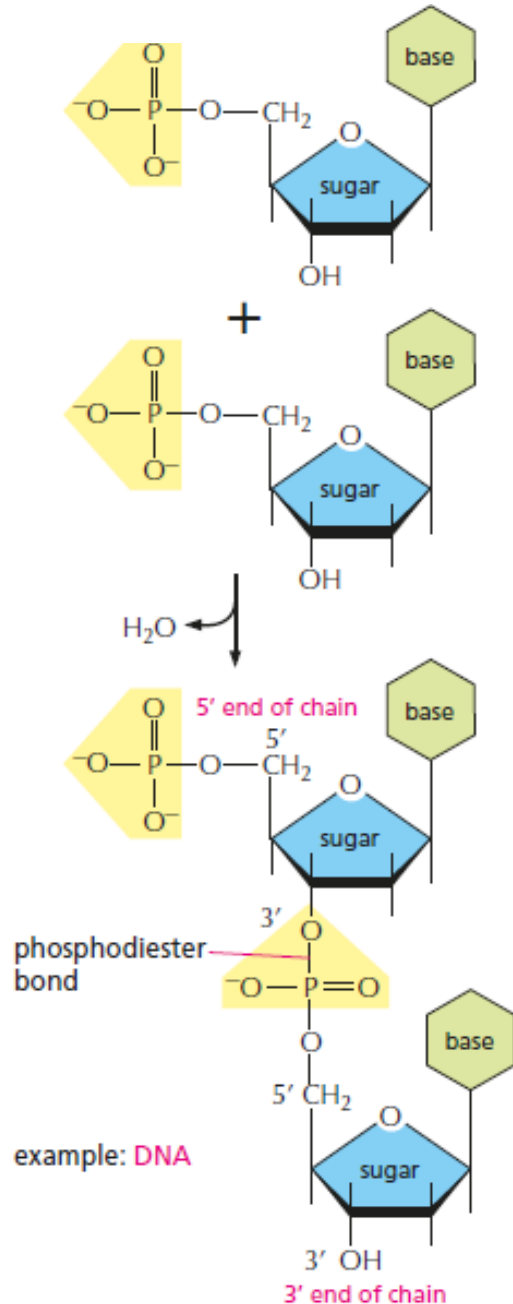
BASE	NUCLEOSIDE	ABBR.
adenine	adenosine	A
guanine	guanosine	G
cytosine	cytidine	C
uracil	uridine	U
thymine	thymidine	T

Nucleotides are abbreviated by three capital letters. Some examples follow:

AMP = adenosine monophosphate
dAMP = deoxyadenosine monophosphate
UDP = uridine diphosphate
ATP = adenosine triphosphate

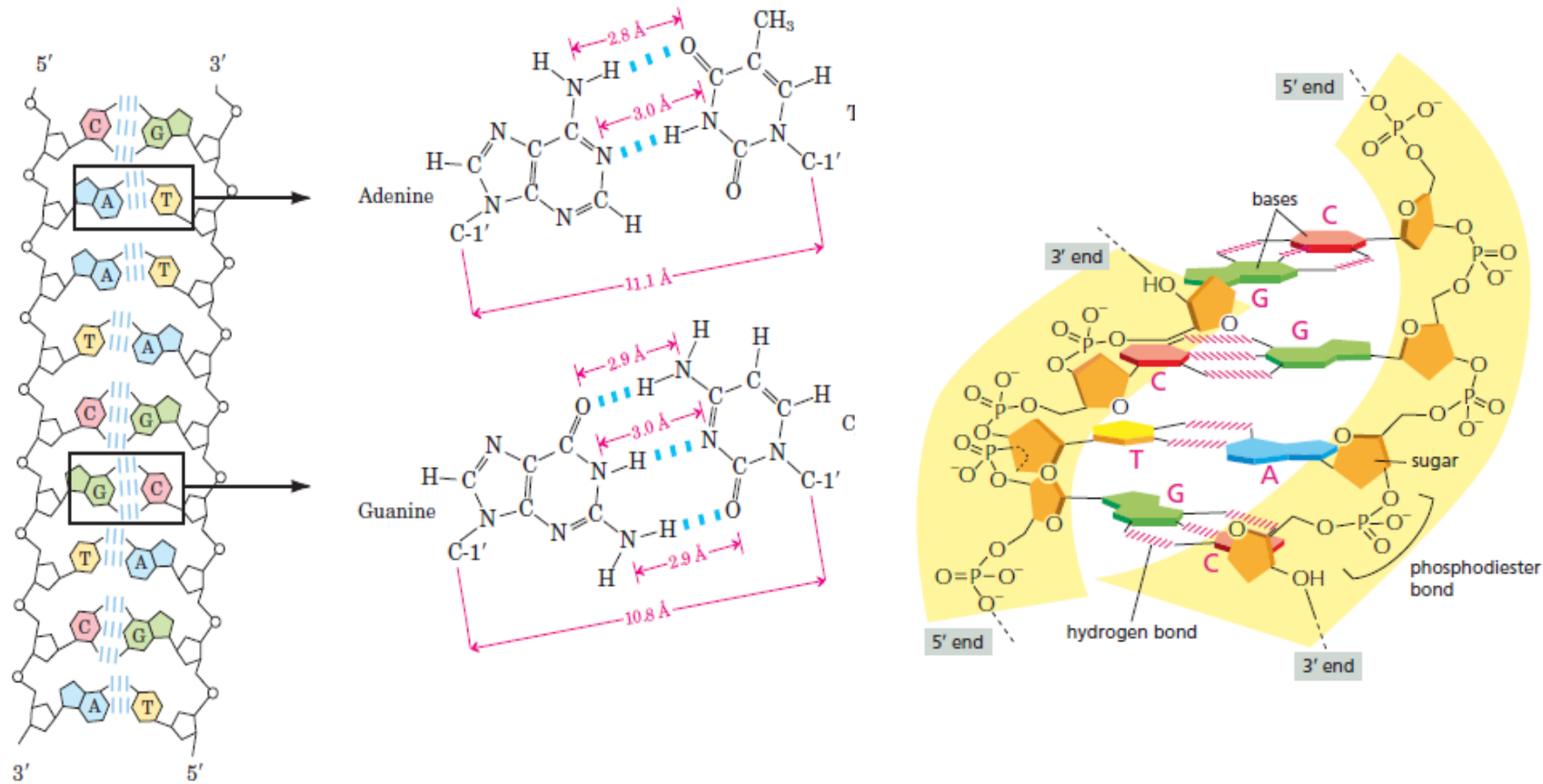


Nucleic Acids



Nucleotides are joined together by phosphodiester bonds between 5' and 3' carbon atoms of the sugar ring, via a phosphate group, to form nucleic acids. The linear sequence of nucleotides in a nucleic acid chain is commonly abbreviated by a one-letter code, such as AGCTTACA, with the 5' end of the chain at the left.

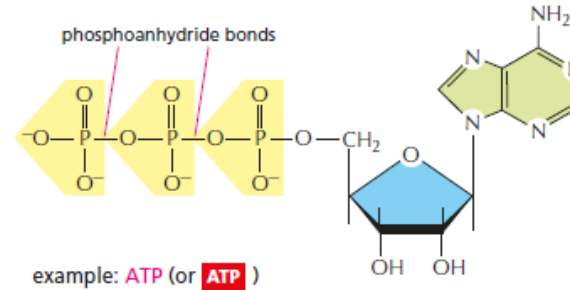
Hydrogen-Bonding Patterns



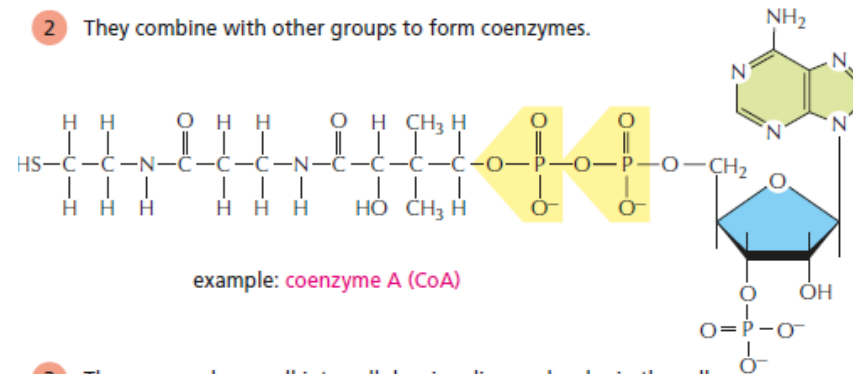
Nucleotides

NUCLEOTIDES HAVE MANY OTHER FUNCTIONS

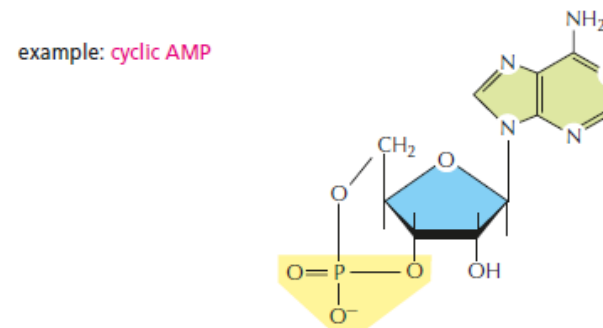
- 1 They carry chemical energy in their easily hydrolyzed phosphoanhydride bonds.



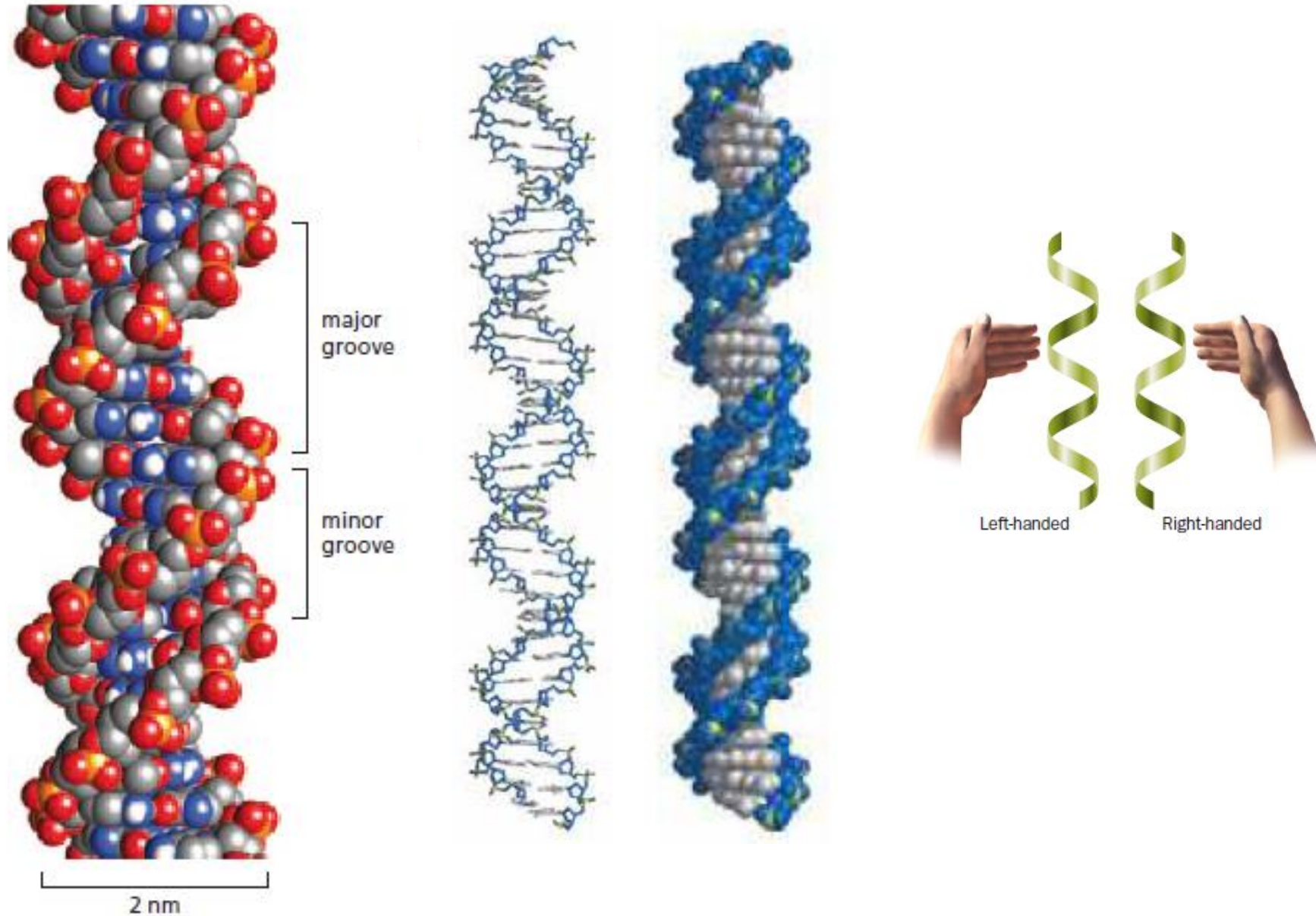
- 2 They combine with other groups to form coenzymes.



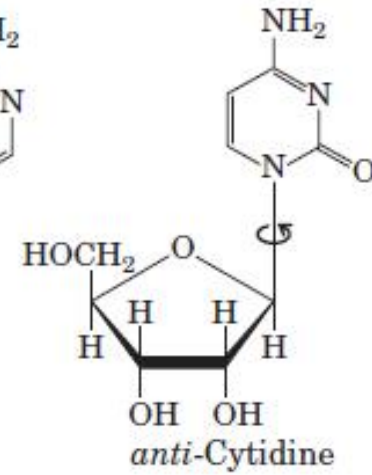
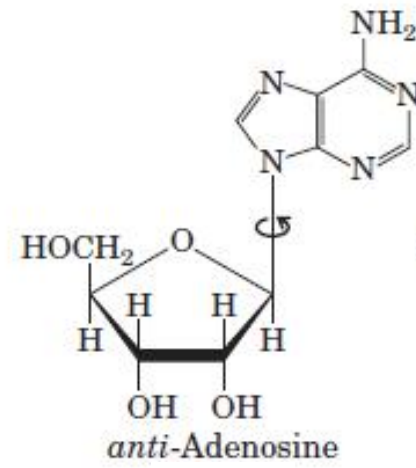
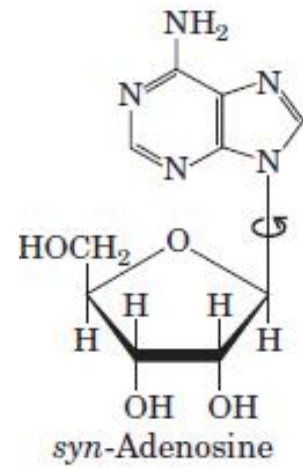
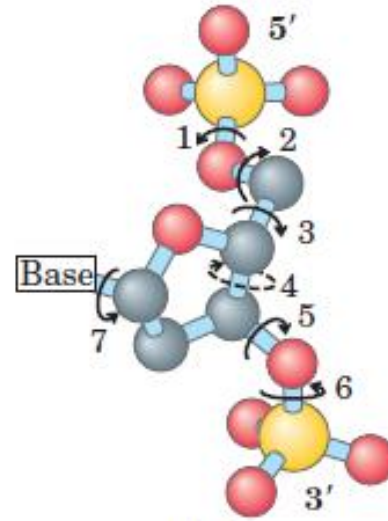
- 3 They are used as small intracellular signaling molecules in the cell.



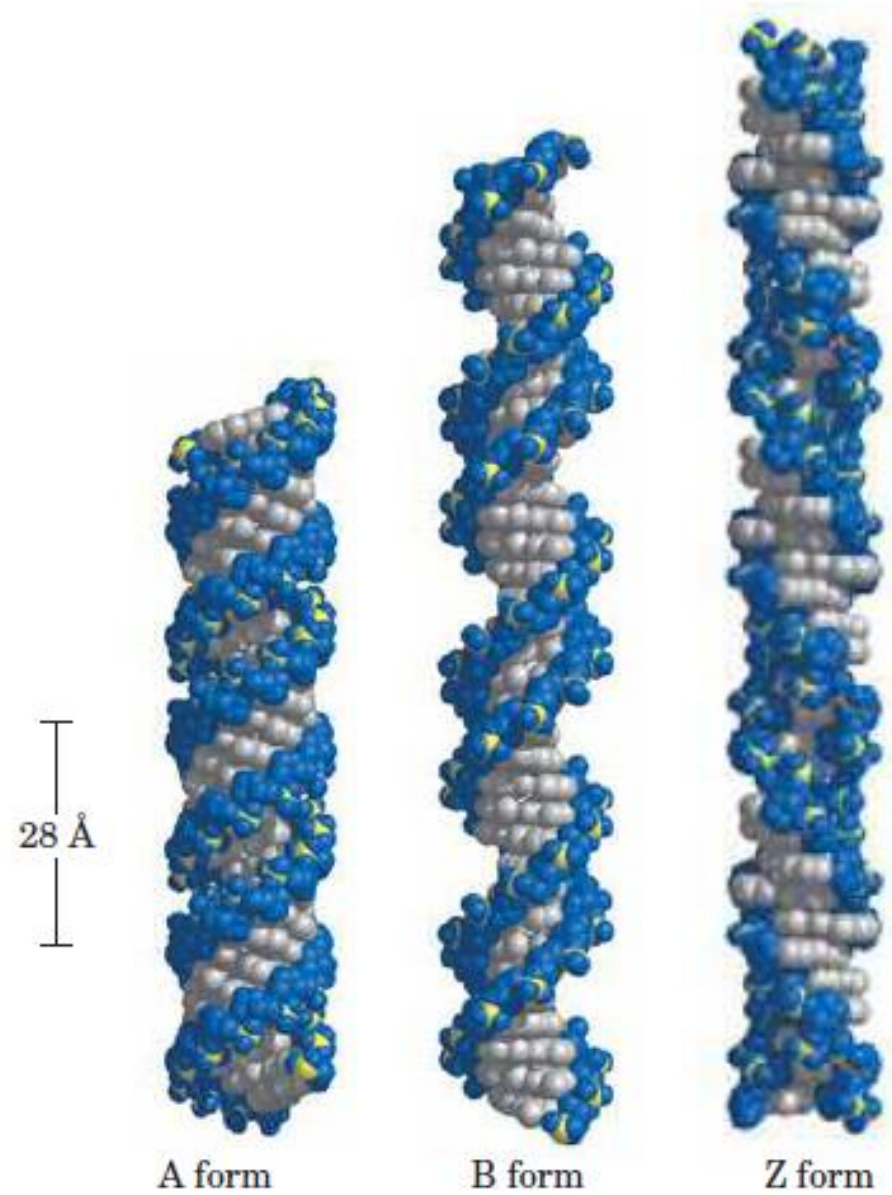
Structure of DNA



Structural variation in DNA

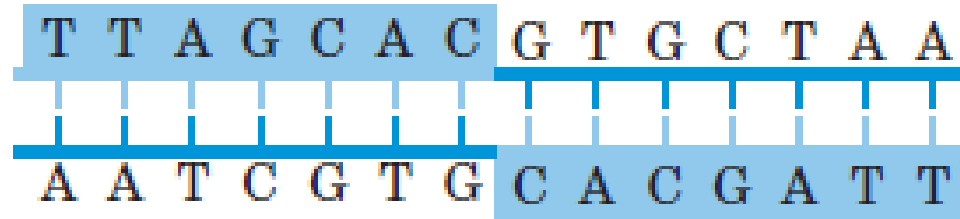
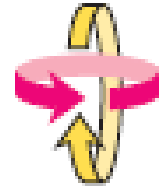


Structural variation in DNA

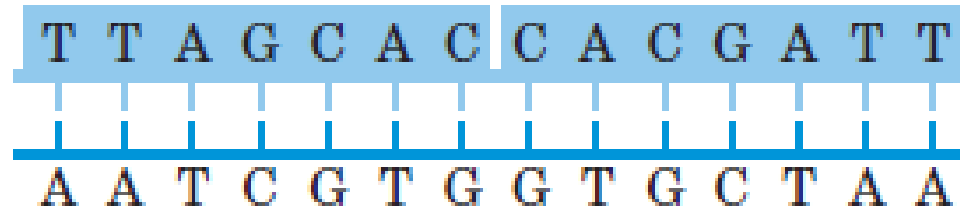


Certain DNA Sequences Adopt Unusual Structures

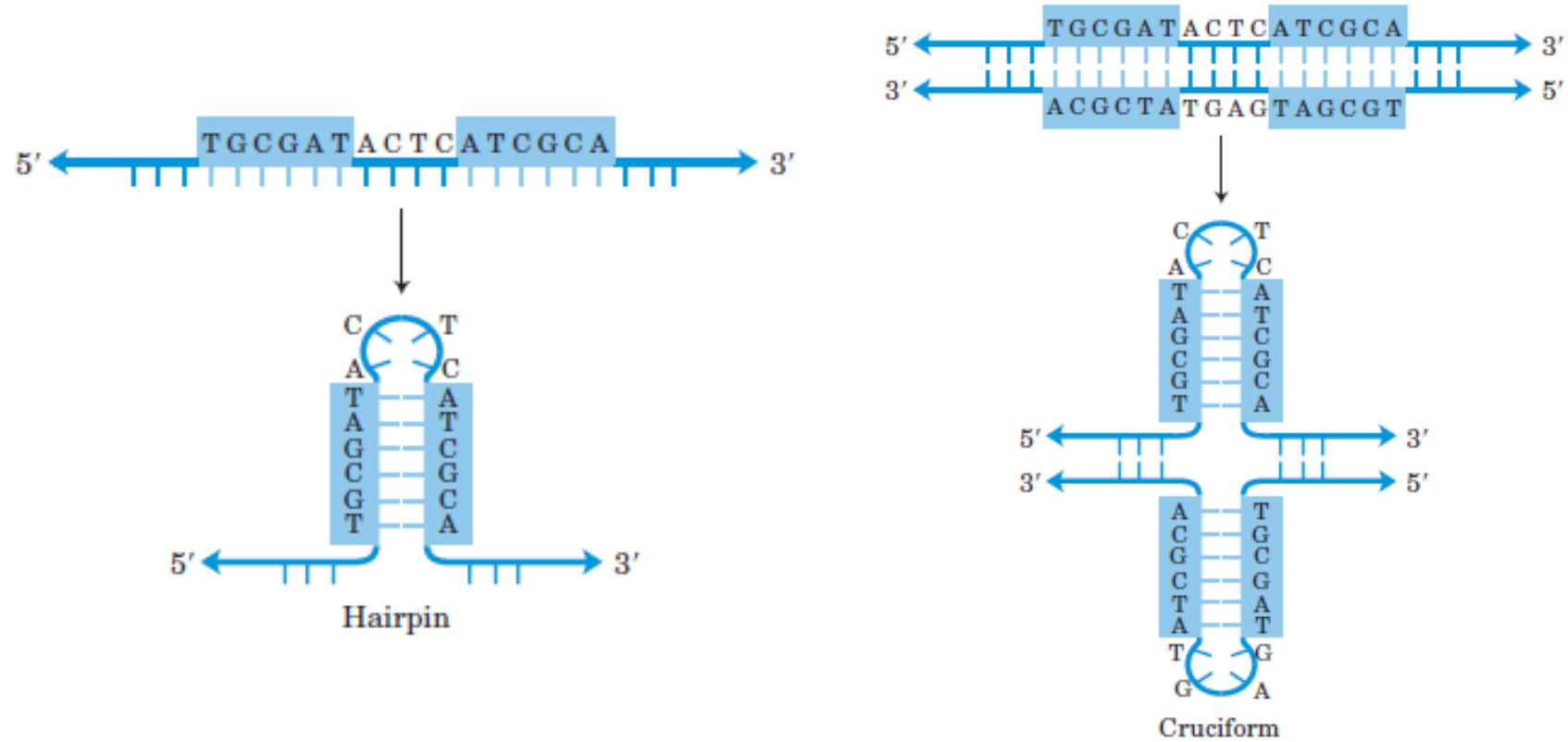
Palindrome



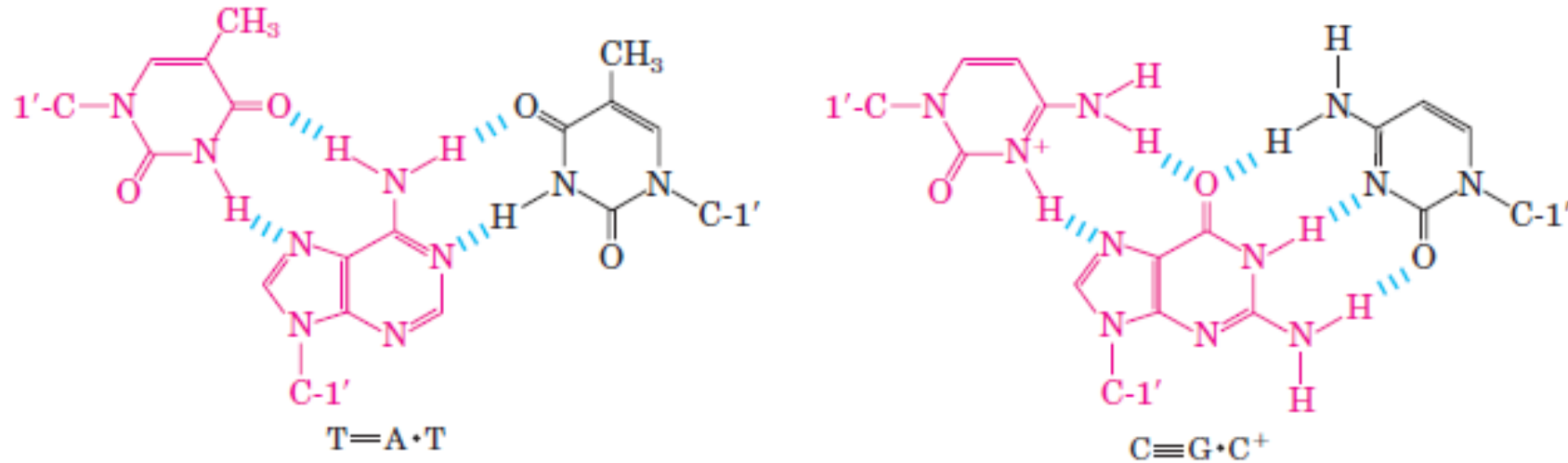
Mirror repeat



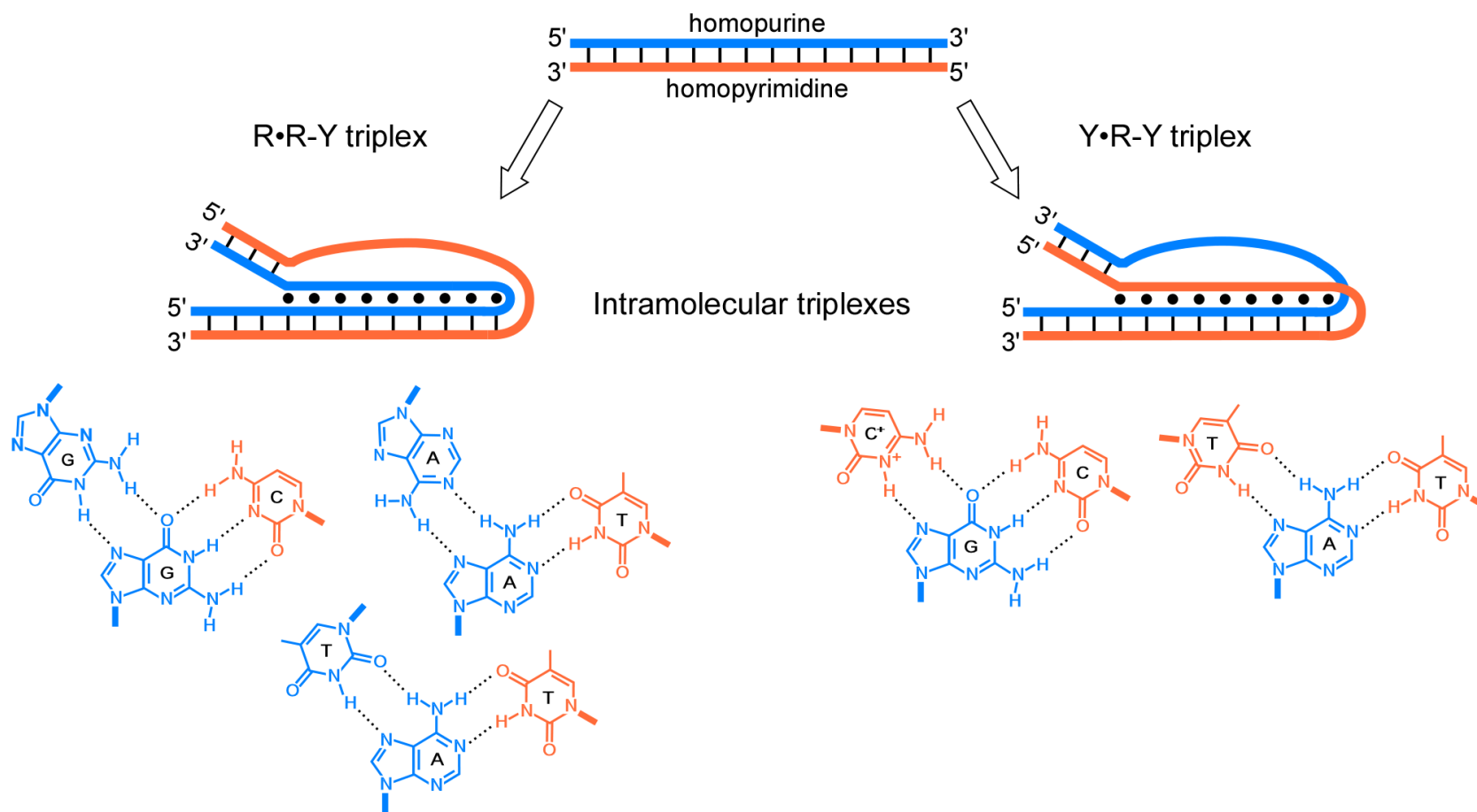
Hairpins and cruciforms

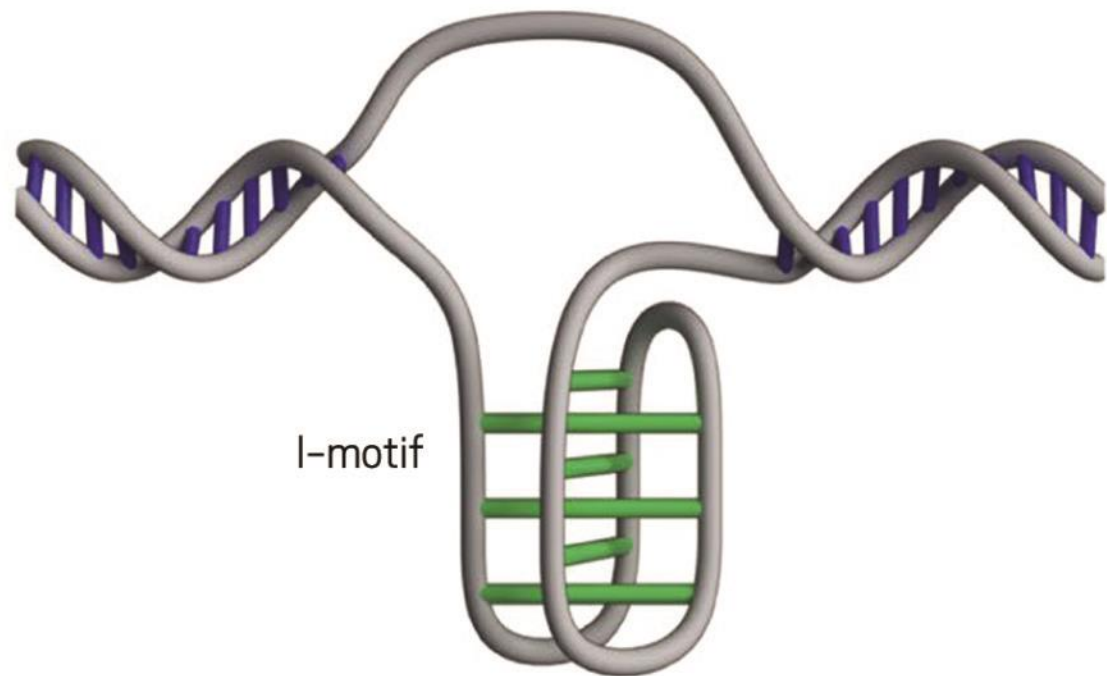
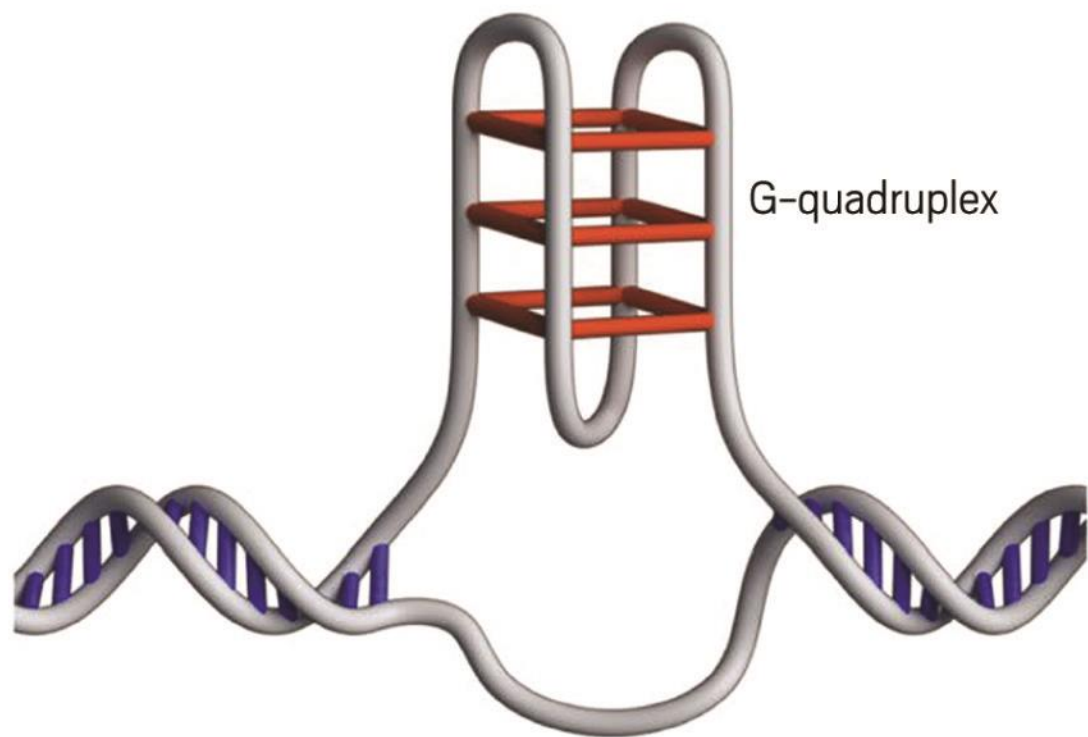


Several unusual DNA structures involve three or even four DNA strands



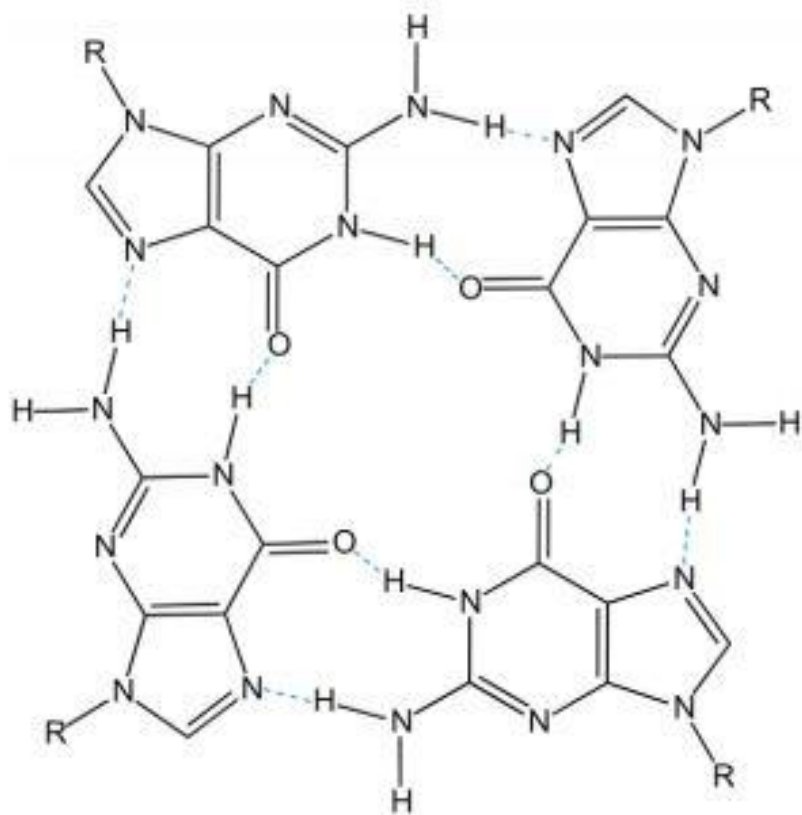
The atoms that participate in the hydrogen bonding of **triplex DNA**, are often referred to as **Hoogsteen positions**, and the non-Watson-Crick pairing is called **Hoogsteen pairing**, after Karst Hoogsteen, who in 1963 first recognized the potential for these unusual pairings



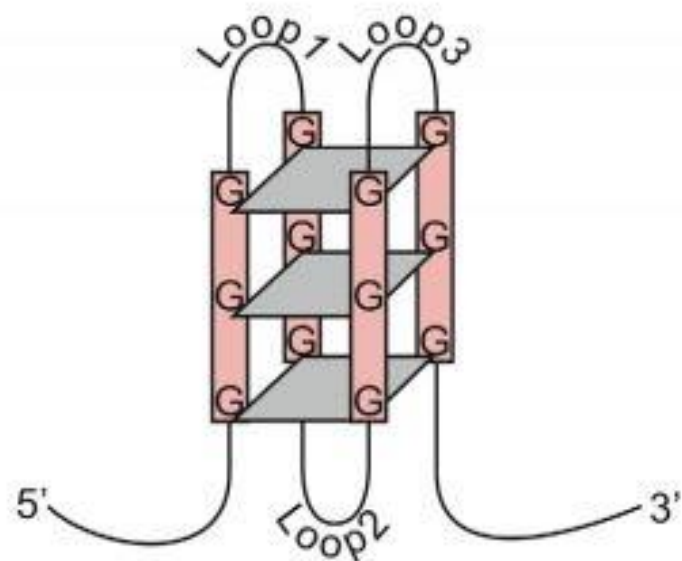


G-quadruplexes

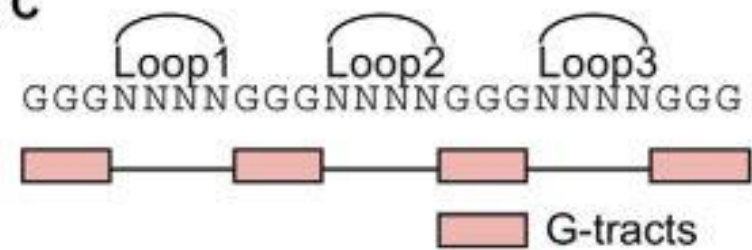
A



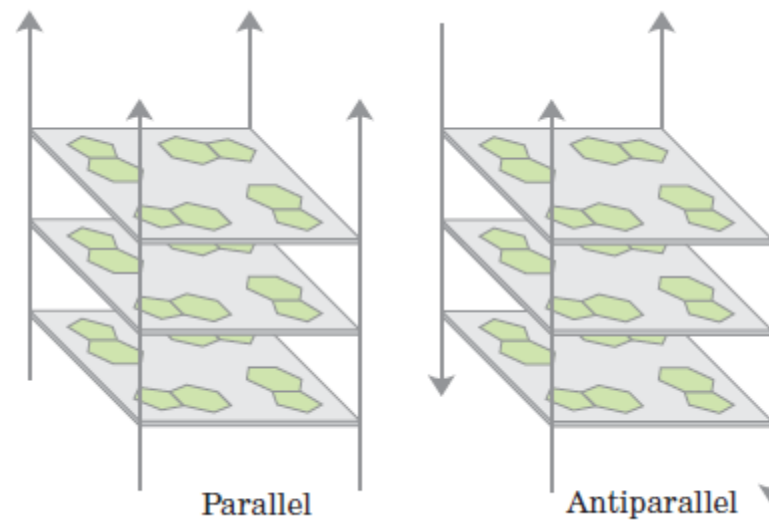
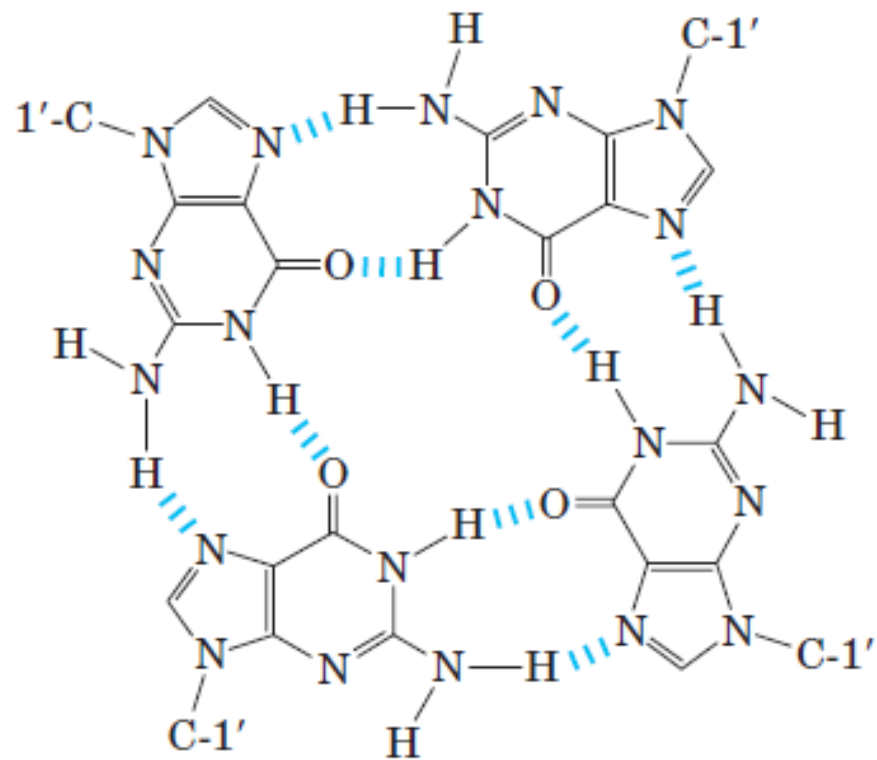
B



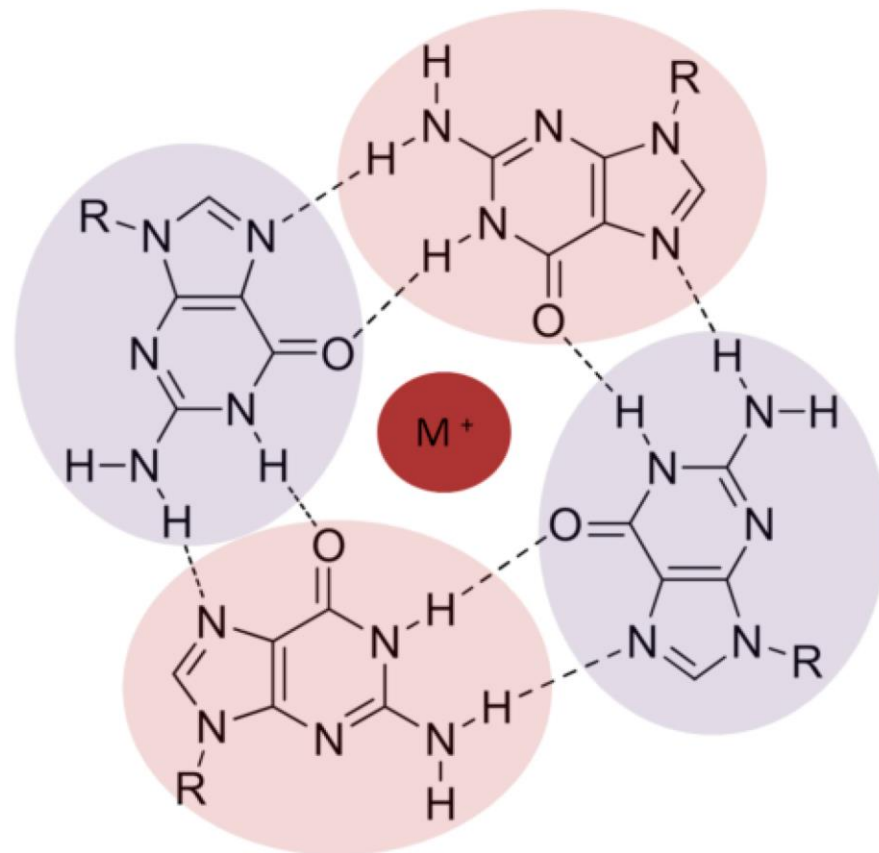
C



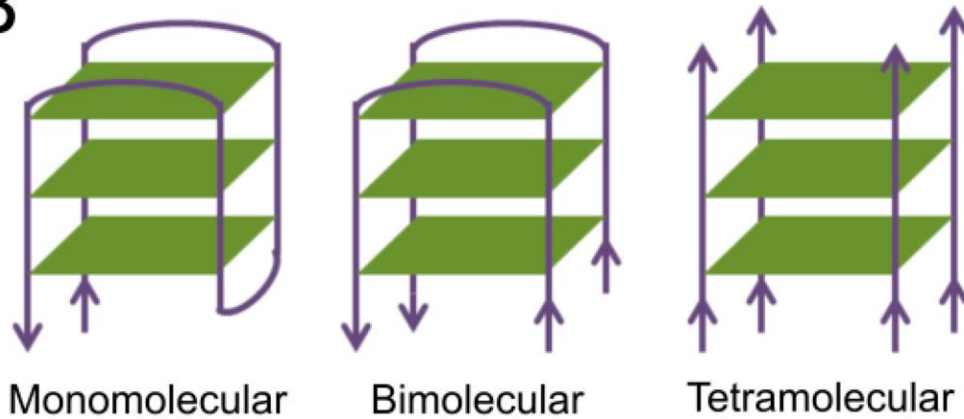
G-quadruplexes

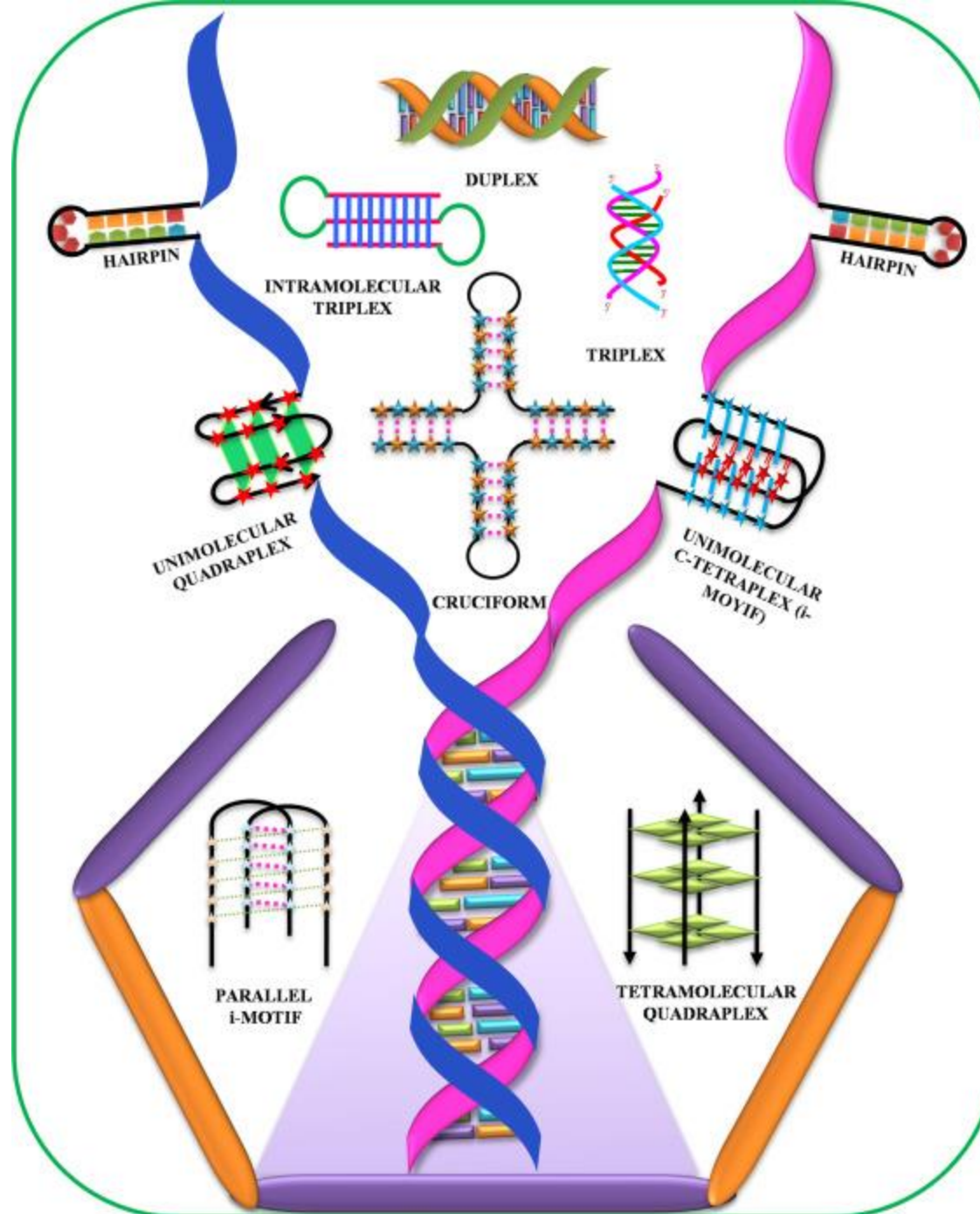


A



B



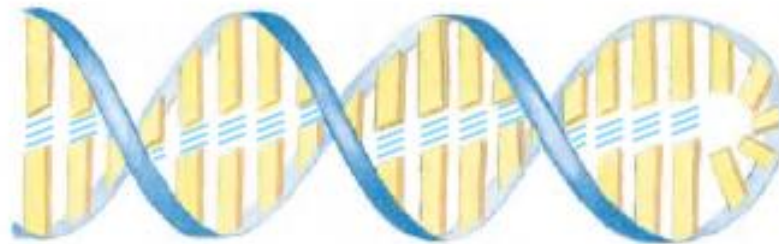
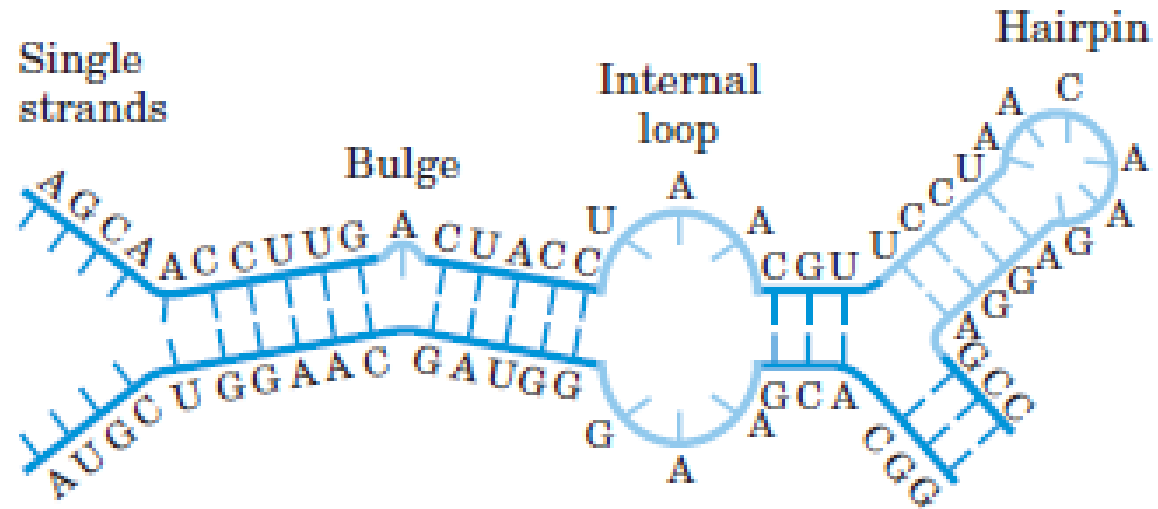


Intramolecular

Vs

Intermolecular

Structure of RNA

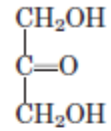


Hairpin double helix

The paired regions generally have an A-form right-handed helix, as shown for a hairpin

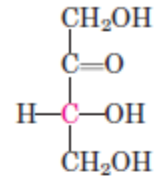
Monosaccharides

Three carbons



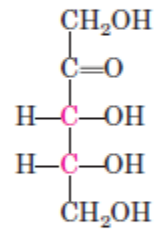
Dihydroxyacetone

Four carbons

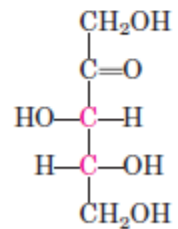


D-Erythrulose

Five carbons

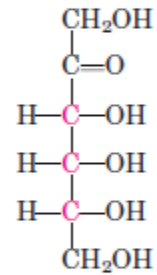


D-Ribulose

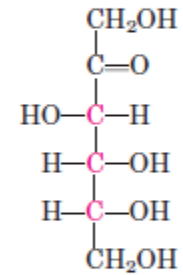


D-Xylulose

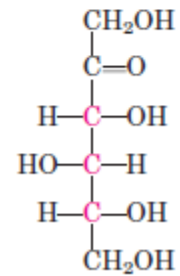
Six carbons



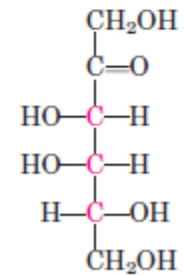
D-Psicose



D-Fructose



D-Sorbose

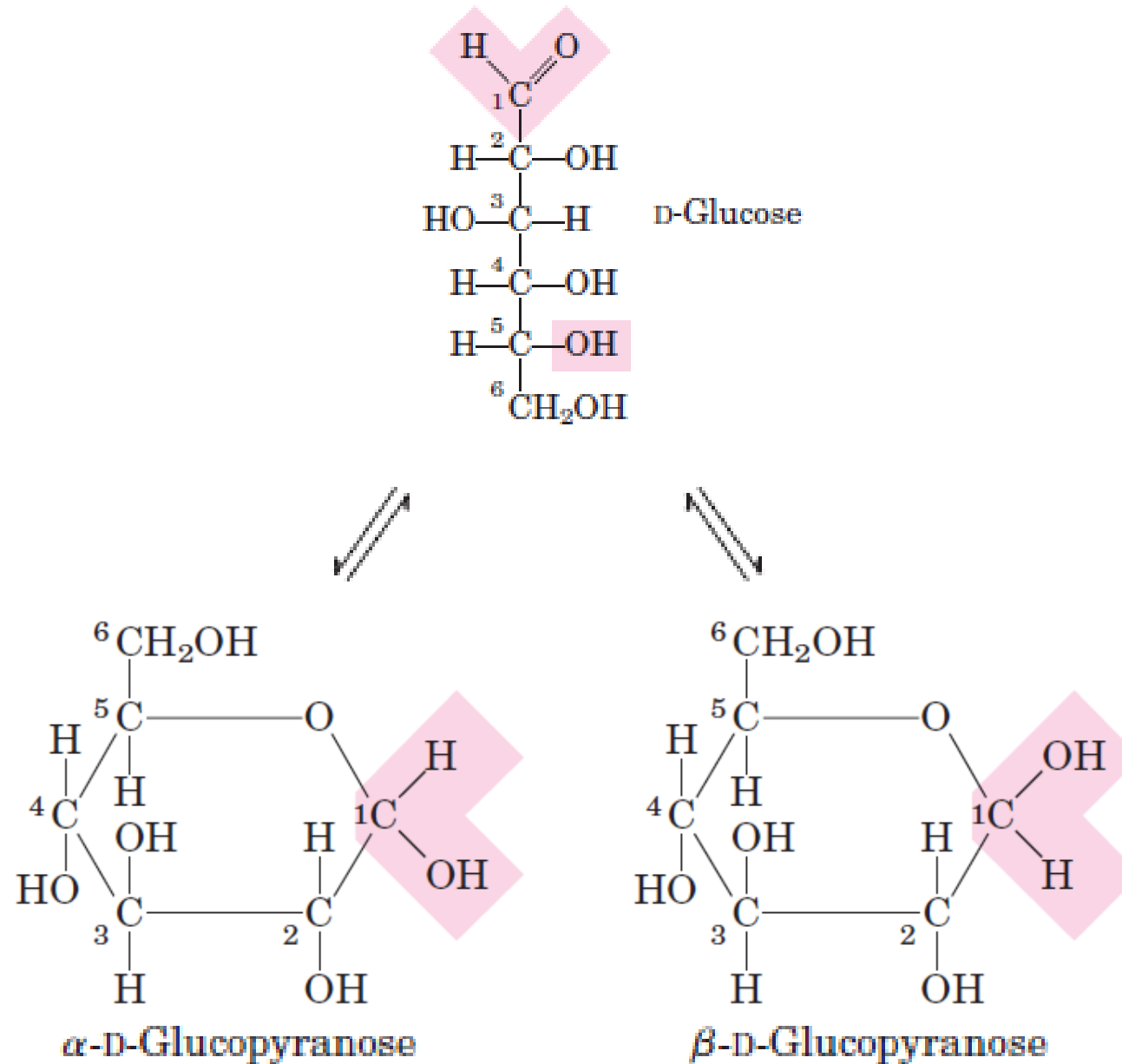


D-Tagatose

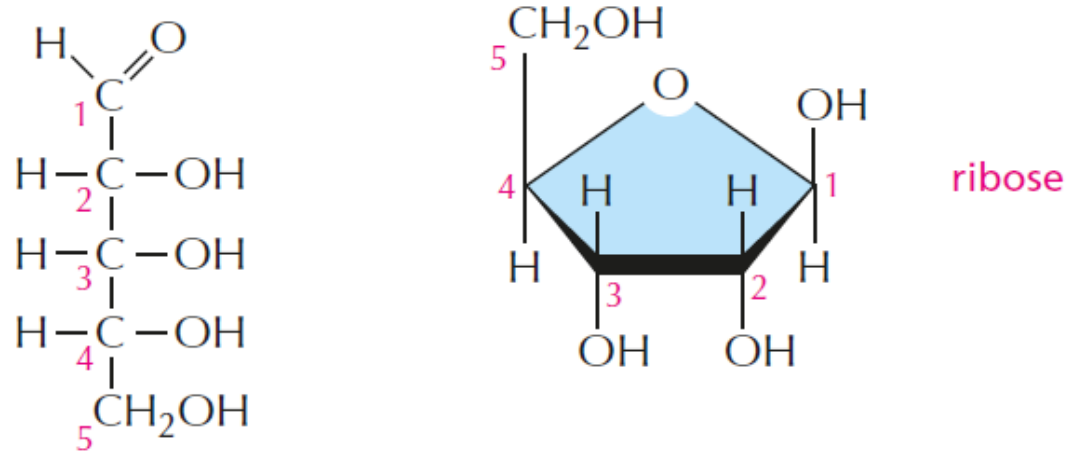
D-Ketoses

Monosaccharides

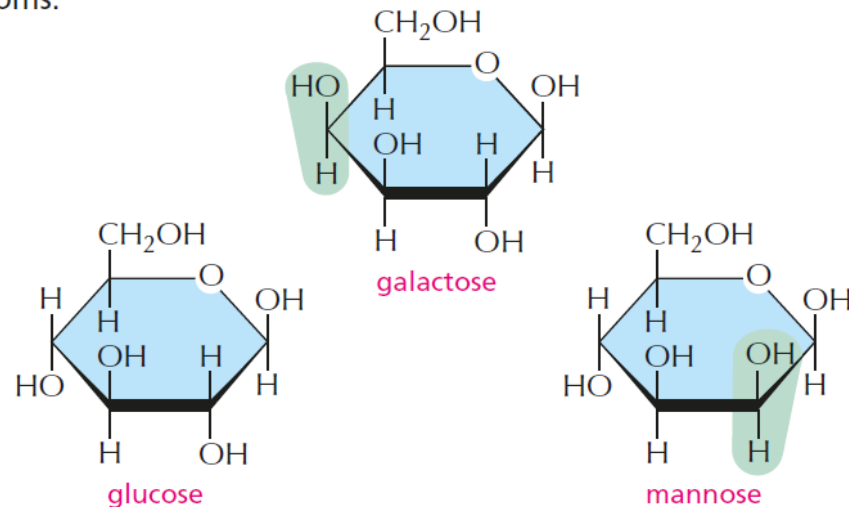
Formation of the two cyclic forms of D-glucose



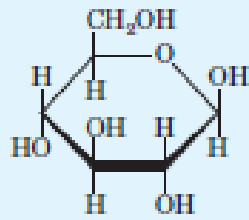
Monosaccharides



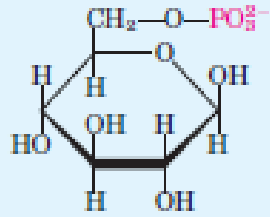
Many monosaccharides differ only in the spatial arrangement of atoms—that is, they are **isomers**. For example, glucose, galactose, and mannose have the same formula ($\text{C}_6\text{H}_{12}\text{O}_6$) but differ in the arrangement of groups around one or two carbon atoms.



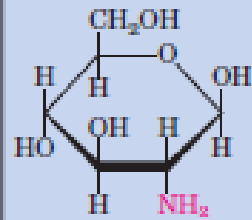
Glucose family



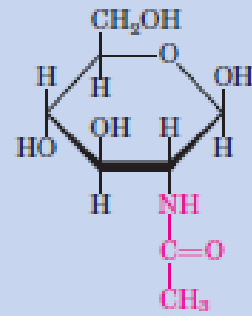
β -D-Glucose



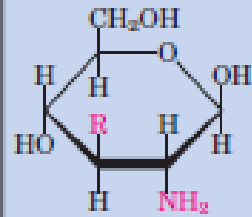
β -D-Glucose 6-phosphate



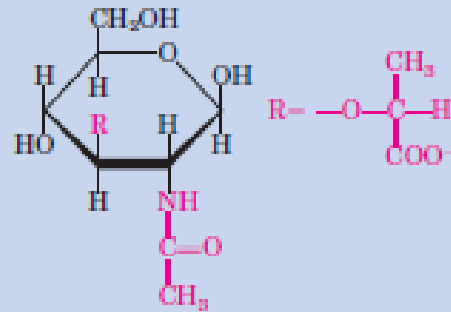
β -D-Glucosamine



N-Acetyl- β -D-glucosamine

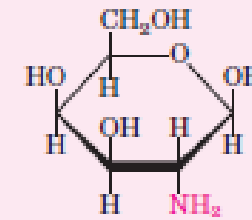


Muramic acid

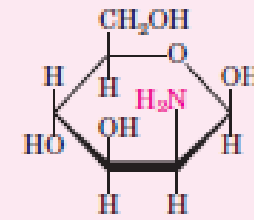


N-Acetylmuramic acid

Amino sugars

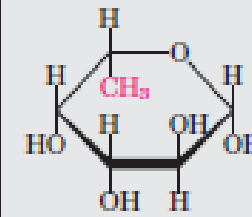


β -D-Galactosamine

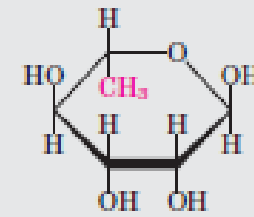


β -D-Mannosamine

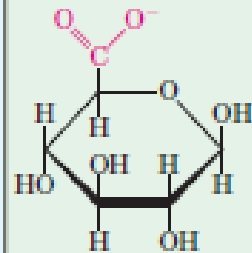
Deoxy sugars



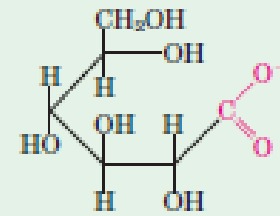
β -L-Fucose



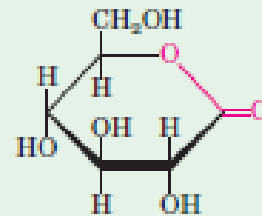
α -L-Rhamnose



β -D-Glucuronate

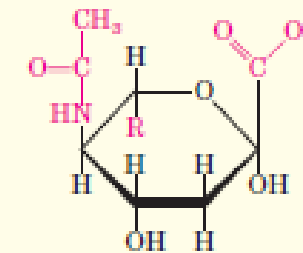


D-Gluconate

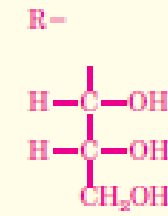


D-Glucono- δ -lactone

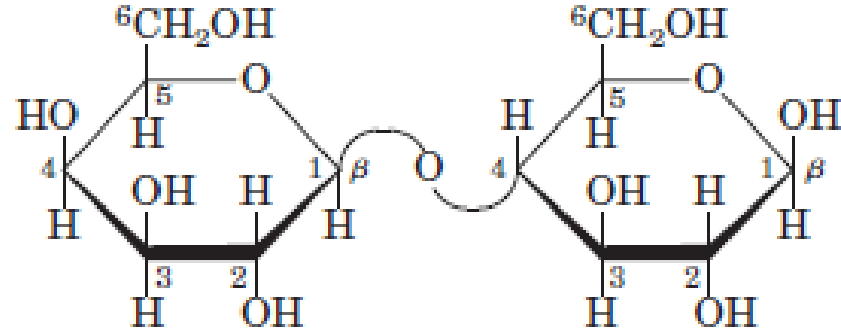
Acidic sugars



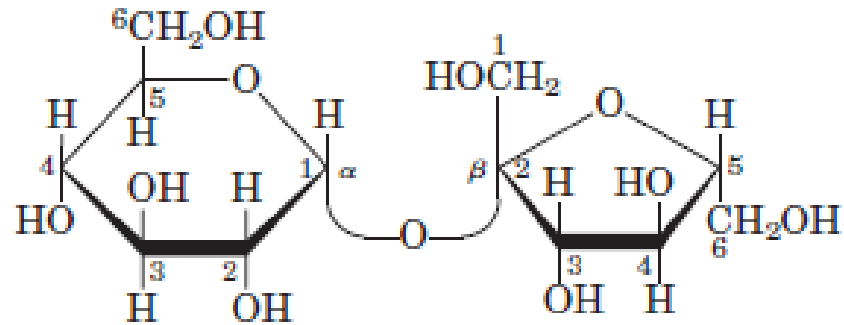
N-Acetylneuraminic acid
(a sialic acid)



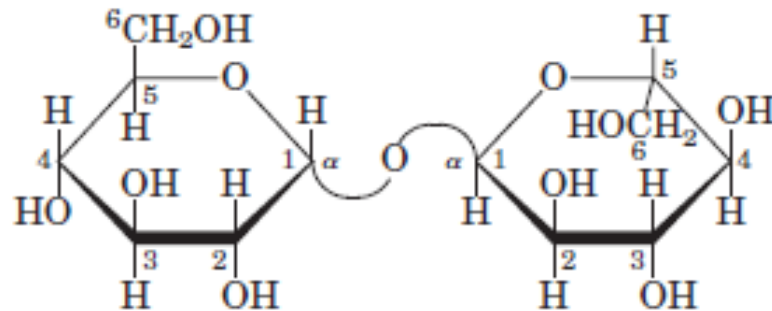
Disaccharides



Lactose (β form)



Sucrose



Trehalose

Three common disaccharides are

maltose (glucose + glucose)

lactose (galactose + glucose)

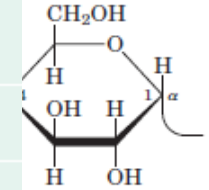
sucrose (glucose + fructose)

Polysaccharides

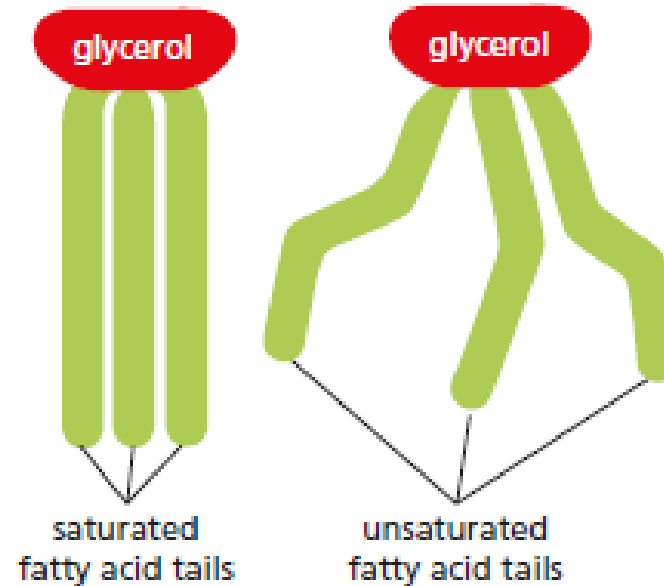
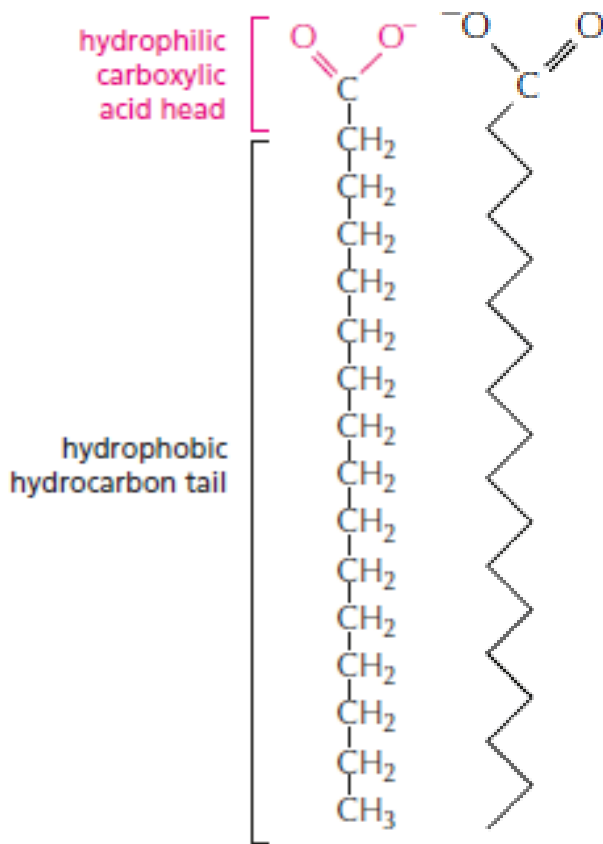
Homopolysaccharides
Unbranched



Polymer	Type*	Repeating unit†	Size (number of monosaccharide units)	Roles/significance
Starch				
Amylose	Homo-	($\alpha 1 \rightarrow 4$)Glc, linear	50–5,000	Energy storage: in plants
Amylopectin	Homo-	($\alpha 1 \rightarrow 4$)Glc, with ($\alpha 1 \rightarrow 6$)Glc branches every 24–30 residues	Up to 10^6	
Glycogen	Homo-	($\alpha 1 \rightarrow 4$)Glc, with ($\alpha 1 \rightarrow 6$)Glc branches every 8–12 residues	Up to 50,000	Energy storage: in bacteria and animal cells
Cellulose	Homo-	($\beta 1 \rightarrow 4$)Glc	Up to 15,000	Structural: in plants, gives rigidity and strength to cell walls
Chitin	Homo-	($\beta 1 \rightarrow 4$)GlcNAc	Very large	Structural: in insects, spiders, crustaceans, gives rigidity and strength to exoskeletons
Dextran	Homo-	($\alpha 1 \rightarrow 6$)Glc, with ($\alpha 1 \rightarrow 3$) branches	Wide range	Structural: in bacteria, extracellular adhesive
Peptidoglycan	Hetero-; peptides attached	4)Mur2Ac($\beta 1 \rightarrow 4$)GlcNAc($\beta 1$	Very large	Structural: in bacteria, gives rigidity and strength to cell envelope
Agarose	Hetero-	3)D-Gal($\beta 1 \rightarrow 4$)3,6-anhydro-L-Gal($\alpha 1$	1,000	Structural: in algae, cell wall material
Hyaluronan (a glycosaminoglycan)	Hetero-; acidic	4)GlcA($\beta 1 \rightarrow 3$)GlcNAc($\beta 1$	Up to 100,000	Structural: in vertebrates, extracellular matrix of skin and connective tissue; viscosity and lubrication in joints



Fats - depend on the length and saturation of the fatty acid chains they carry

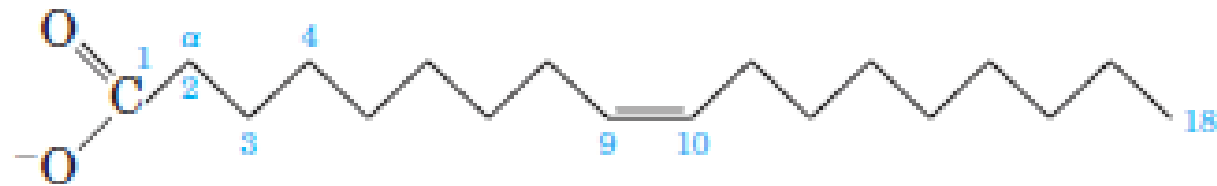


Fatty acids are stored in the cytoplasm of many cells in the form of droplets of *triacylglycerol* molecules

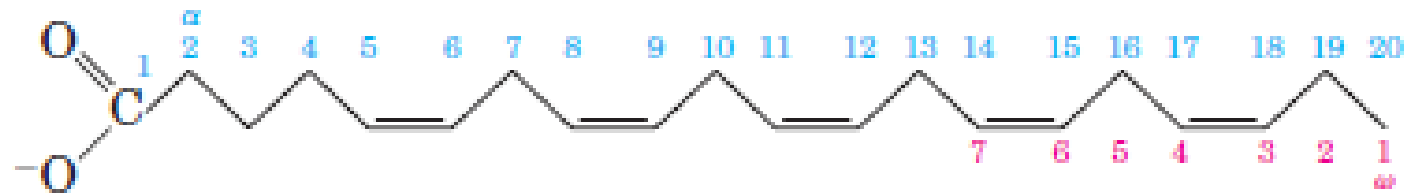
Saturated fats are found in meat and dairy products

Plant oils, such as corn oil, contain unsaturated fatty acids

Fatty acids are carboxylic acids with hydrocarbon chains ranging from 4 to 36 carbons long (C₄ to C₃₆).



(a) 18:1(Δ^9) *cis*-9-Octadecenoic acid



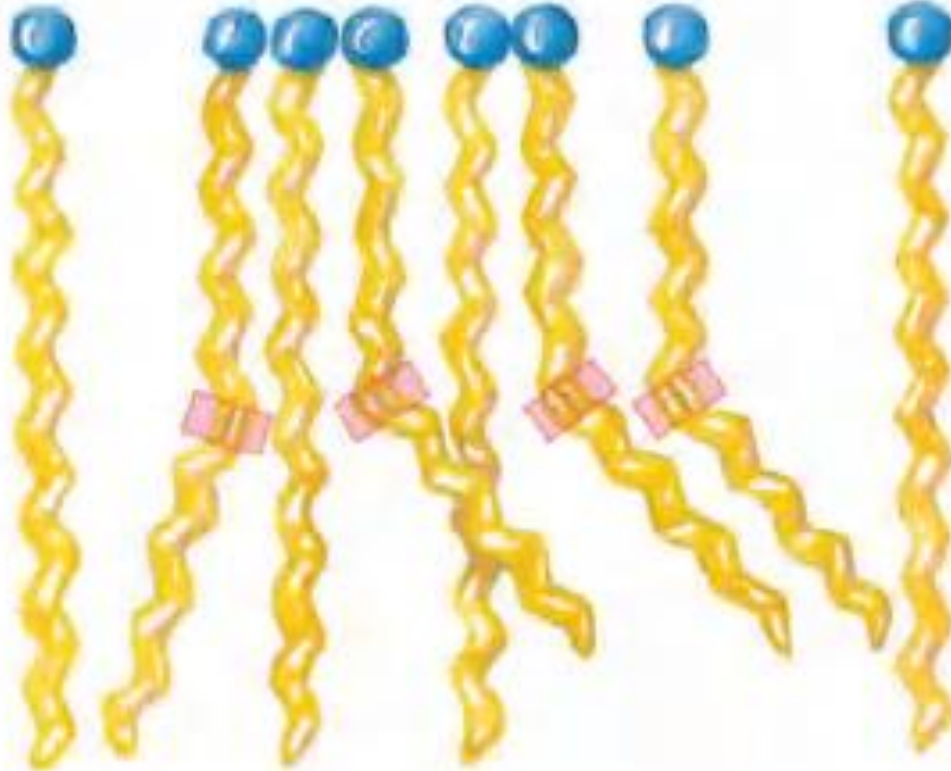
(b) 20:5($\Delta^{5,8,11,14,17}$) Eicosapentaenoic acid (EPA),
an omega-3 fatty acid

Carbon skeleton	Structure*	Systematic name [†]	Common name (derivation)
12:0	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	<i>n</i> -Dodecanoic acid	Lauric acid (Latin <i>laurus</i> , "laurel plant")
14:0	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	<i>n</i> -Tetradecanoic acid	Myristic acid (Latin <i>Myristica</i> , nutmeg genus)
16:0	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	<i>n</i> -Hexadecanoic acid	Palmitic acid (Latin <i>palma</i> , "palm tree")
18:0	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	<i>n</i> -Octadecanoic acid	Stearic acid (Greek <i>stear</i> , "hard fat")
20:0	$\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$	<i>n</i> -Eicosanoic acid	Arachidic acid (Latin <i>Arachis</i> , legume genus)
24:0	$\text{CH}_3(\text{CH}_2)_{22}\text{COOH}$	<i>n</i> -Tetracosanoic acid	Lignoceric acid (Latin <i>lignum</i> , "wood" + <i>cera</i> , "wax")
16:1(Δ^9)	$\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -9-Hexadecenoic acid	Palmitoleic acid
18:1(Δ^9)	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -9-Octadecenoic acid	Oleic acid (Latin <i>oleum</i> , "oil")
18:2($\Delta^{9,12}$)	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -, <i>cis</i> -9,12-Octadecadienoic acid	Linoleic acid (Greek <i>linon</i> , "flax")
18:3($\Delta^{9,12,15}$)	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -, <i>cis</i> -, <i>cis</i> -9,12,15-Octadecatrienoic acid	α -Linolenic acid
20:4($\Delta^{5,8,11,14}$)	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_3\text{COOH}$	<i>cis</i> -, <i>cis</i> -, <i>cis</i> -, <i>cis</i> -5, 8, 11, 14-Icosatetraenoic acid	Arachidonic acid

The packing of fatty acids into stable aggregates

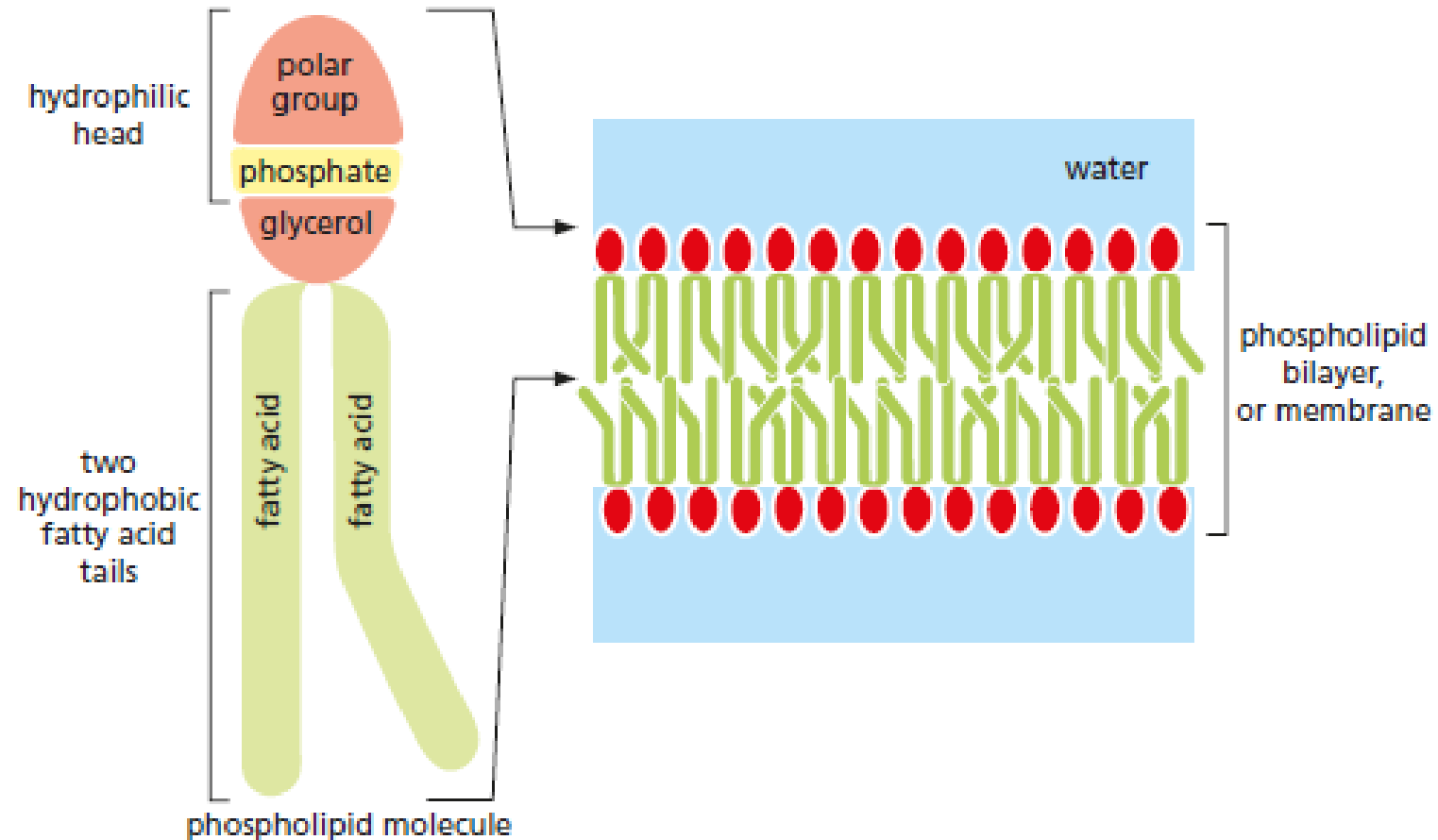


Saturated
fatty acids



Mixture of saturated and
unsaturated fatty acids

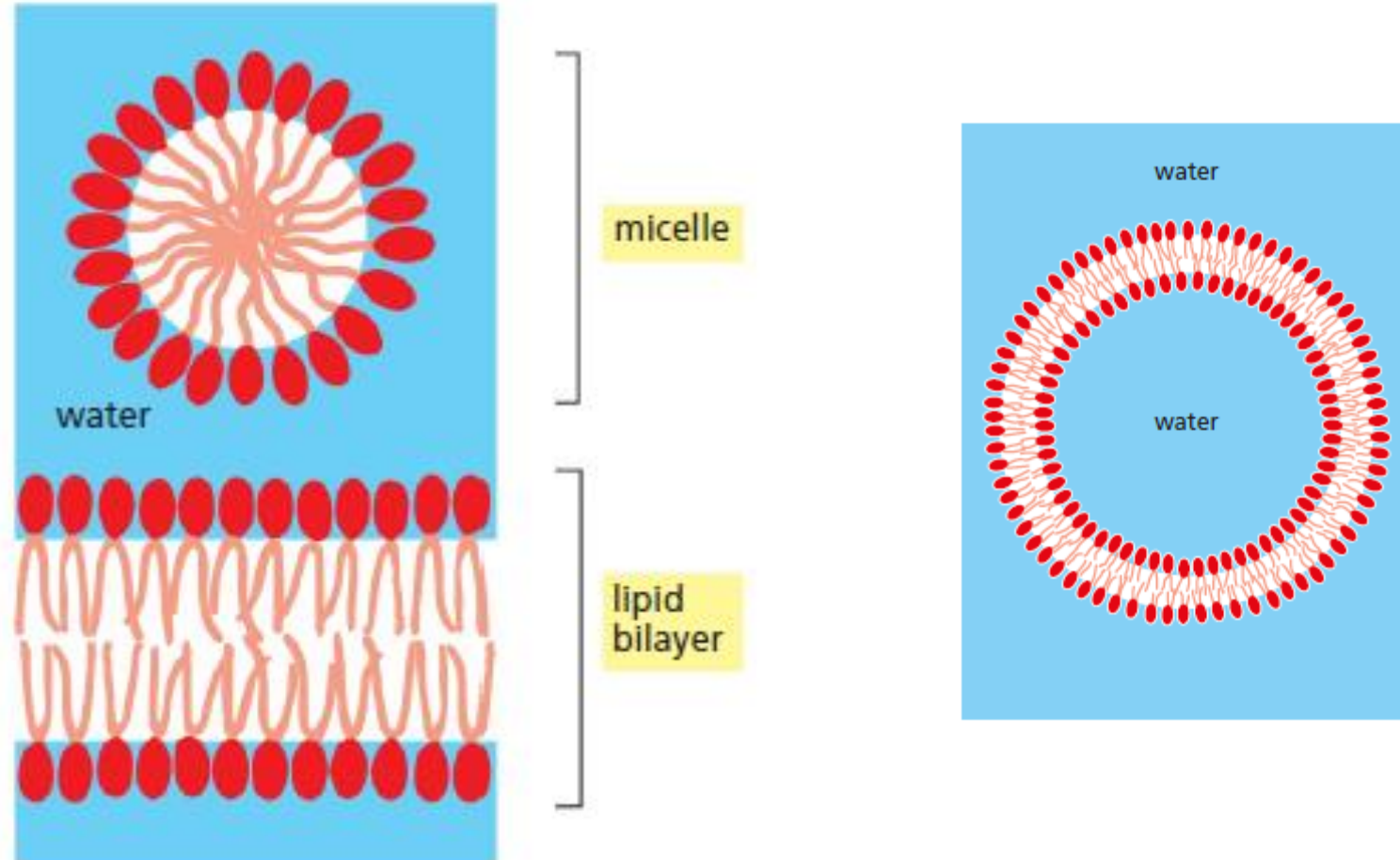
Phospholipid - Lipid Bilayer



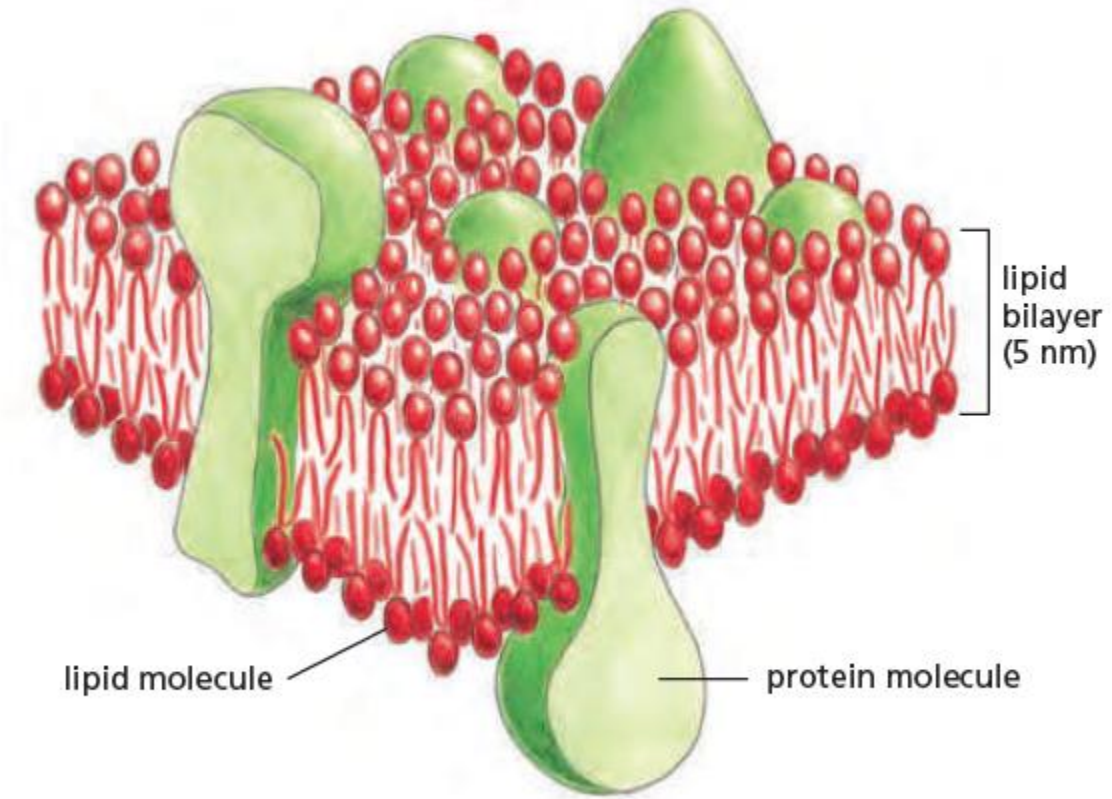
Glycerol is joined to two fatty acid chains.

The remaining –OH group on the glycerol is linked to a hydrophilic phosphate group, which in turn is attached to a small hydrophilic compound such as choline.

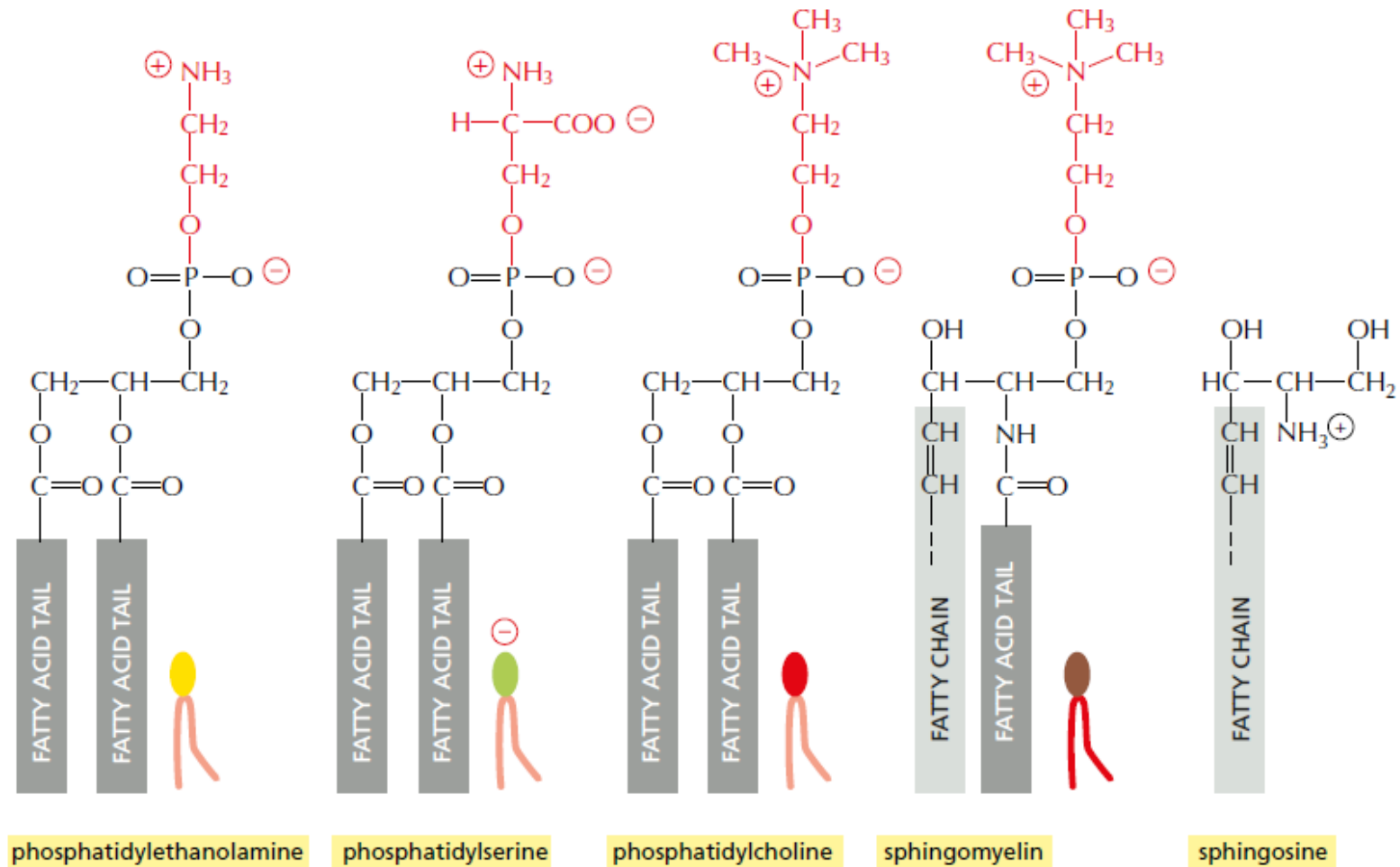
Phospholipids Spontaneously Form Bilayers



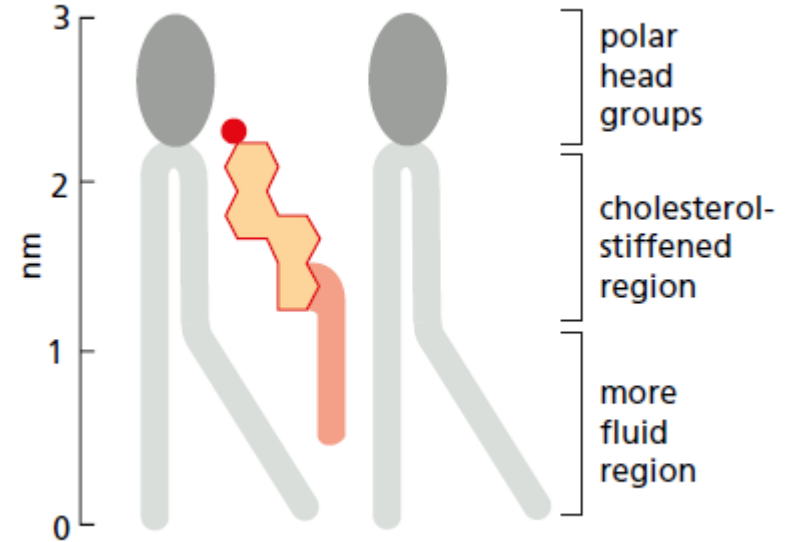
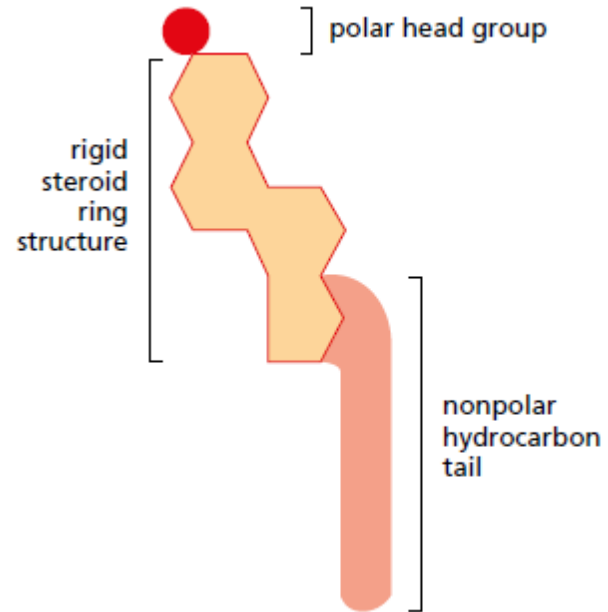
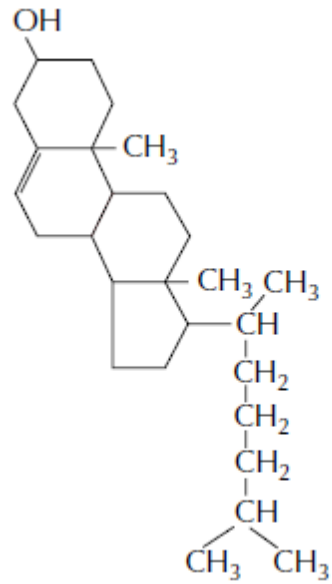
A three-dimensional schematic view of a cell
membrane



Four major phospholipids in mammalian plasma membranes



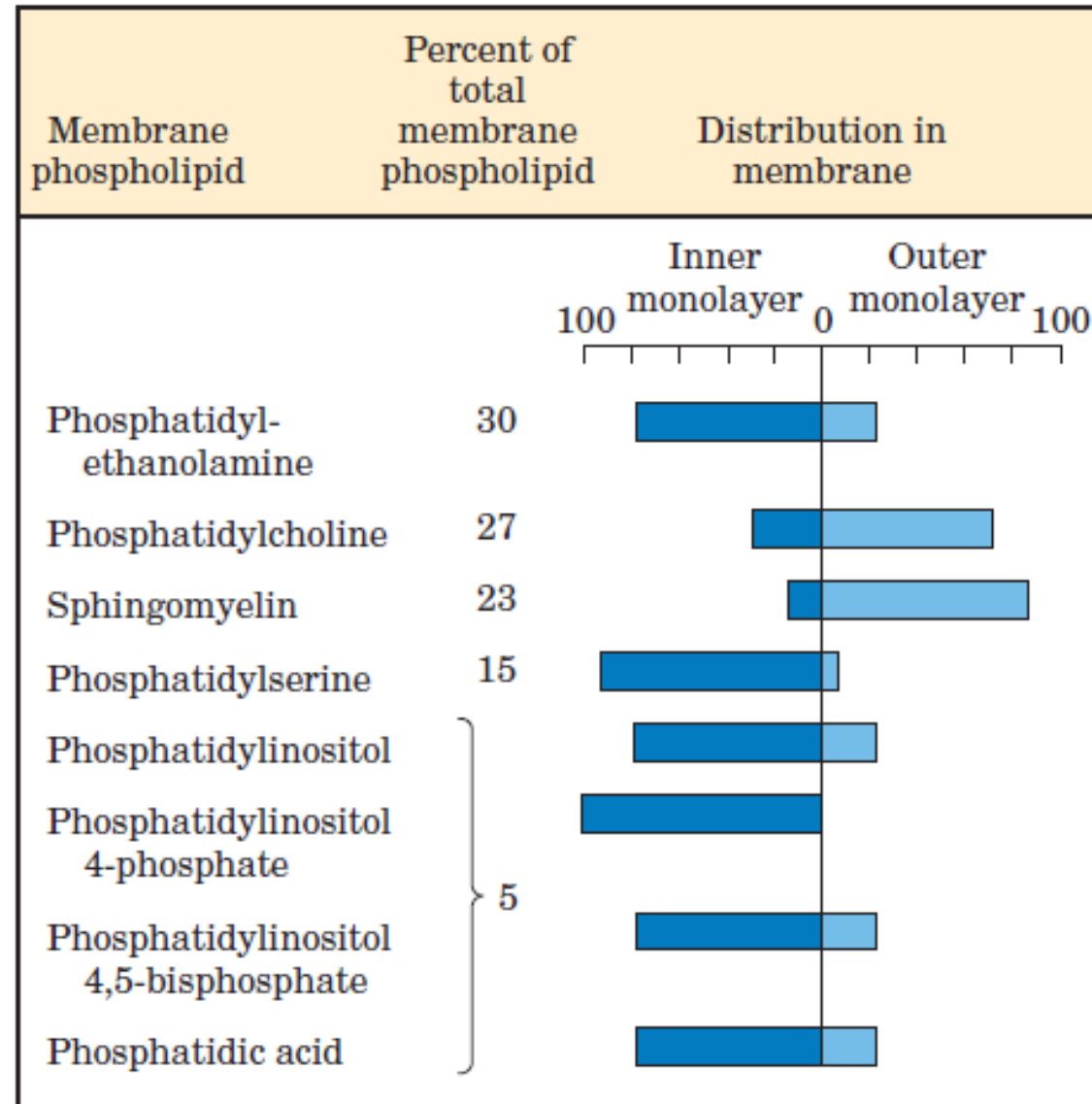
The structure of cholesterol



Lipid composition of the plasma membrane and organelle membranes of a hepatocyte

Lipid	Percentage of total lipid by weight					
	Liver cell plasma membrane	Red blood cell plasma membrane	Myelin	Mitochondrion (inner and outer membranes)	Endoplasmic reticulum	<i>E. coli</i> bacterium
Cholesterol	17	23	22	3	6	0
Phosphatidylethanolamine	7	18	15	28	17	70
Phosphatidylserine	4	7	9	2	5	trace
Phosphatidylcholine	24	17	10	44	40	0
Sphingomyelin	19	18	8	0	5	0
Glycolipids	7	3	28	trace	trace	0
Others	22	14	8	23	27	30

Distribution of phospholipids between the inner and outer monolayers



Proteins

ENZYMES

function: Catalyze covalent bond breakage or formation.



examples: Living cells contain thousands of different enzymes, each of which catalyzes (speeds up) one particular reaction. Examples include: *tryptophan synthetase*—makes the amino acid tryptophan; *pepsin*—degrades dietary proteins in the stomach; *ribulose biphosphate carboxylase*—helps convert carbon dioxide into sugars in plants; *DNA polymerase*—copies DNA; *protein kinase*—adds a phosphate group to a protein molecule.

STRUCTURAL PROTEINS

function: Provide mechanical support to cells and tissues.



examples: Outside cells, *collagen* and *elastin* are common constituents of extracellular matrix and form fibers in tendons and ligaments. Inside cells, *tubulin* forms long, stiff microtubules, and *actin* forms filaments that underlie and support the plasma membrane; *keratin* forms fibers that reinforce epithelial cells and is the major protein in hair and horn.

TRANSPORT PROTEINS

function: Carry small molecules or ions.



examples: In the bloodstream, *serum albumin* carries lipids, *hemoglobin* carries oxygen, and *transferrin* carries iron. Many proteins embedded in cell membranes transport ions or small molecules across the membrane. For example, the bacterial protein *bacteriorhodopsin* is a light-activated proton pump that transports H^+ ions out of the cell; *glucose carriers* shuttle glucose into and out of cells; and a Ca^{2+} pump clears Ca^{2+} from a muscle cell's cytosol after the ions have triggered a contraction.

MOTOR PROTEINS

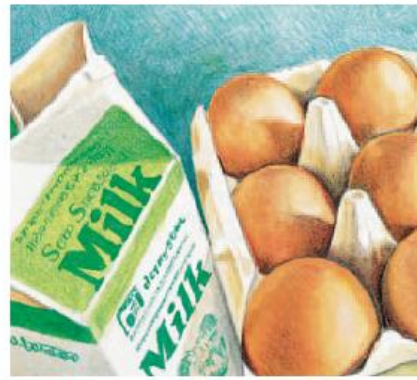
function: Generate movement in cells and tissues.



examples: *Myosin* in skeletal muscle cells provides the motive force for humans to move; *kinesin* interacts with microtubules to move organelles around the cell; *dynein* enables eukaryotic cilia and flagella to beat.

STORAGE PROTEINS

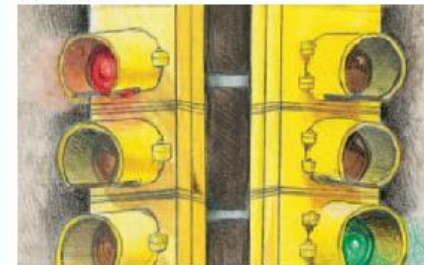
function: Store amino acids or ions.



examples: Iron is stored in the liver by binding to the small protein *ferritin*; *ovalbumin* in egg white is used as a source of amino acids for the developing bird embryo; *casein* in milk is a source of amino acids for baby mammals.

SIGNAL PROTEINS

function: Carry extracellular signals from cell to cell.



examples: Many of the hormones and growth factors that coordinate physiological functions in animals are proteins; *insulin*, for example, is a small protein that controls glucose levels in the blood; *netrin* attracts growing nerve cell axons to specific locations in the developing spinal cord; *nerve growth factor (NGF)* stimulates some types of nerve cells to grow axons; *epidermal growth factor (EGF)* stimulates the growth and division of epithelial cells.

RECEPTOR PROTEINS

function: Detect signals and transmit them to the cell's response machinery.



examples: *Rhodopsin* in the retina detects light; the *acetylcholine receptor* in the membrane of a muscle cell is activated by acetylcholine released from a nerve ending; the *insulin receptor* allows a cell to respond to the hormone insulin by taking up glucose; the *adrenergic receptor* on heart muscle increases the rate of the heartbeat when it binds to adrenaline.

GENE REGULATORY PROTEINS

function: Bind to DNA to switch genes on or off.



examples: The *lactose repressor* in bacteria silences the genes for the enzymes that degrade the sugar lactose; many different *homeodomain proteins* act as genetic switches to control development in multicellular organisms, including humans.

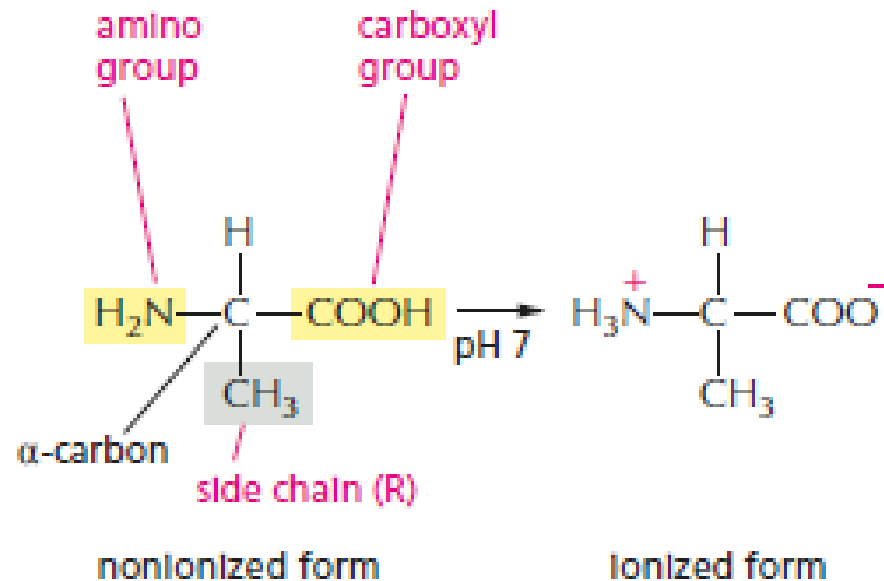
SPECIAL-PURPOSE PROTEINS

function: Highly variable.



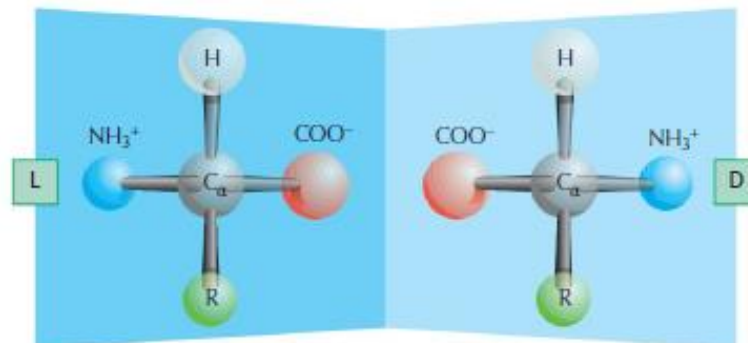
examples: Organisms make many proteins with highly specialized properties. These molecules illustrate the amazing range of functions that proteins can perform. The *antifreeze proteins* of Arctic and Antarctic fishes protect their blood against freezing; *green fluorescent protein* from jellyfish emits a green light; *monellin*, a protein found in an African plant, has an intensely sweet taste; mussels and other marine organisms secrete *glue proteins* that attach them firmly to rocks, even when immersed in seawater.

Amino acids



OPTICAL ISOMERS

The α -carbon atom is asymmetric, which allows for two mirror images (or stereo-) isomers, L and D.



Proteins consist exclusively of L-amino acids.

FAMILIES OF AMINO ACIDS

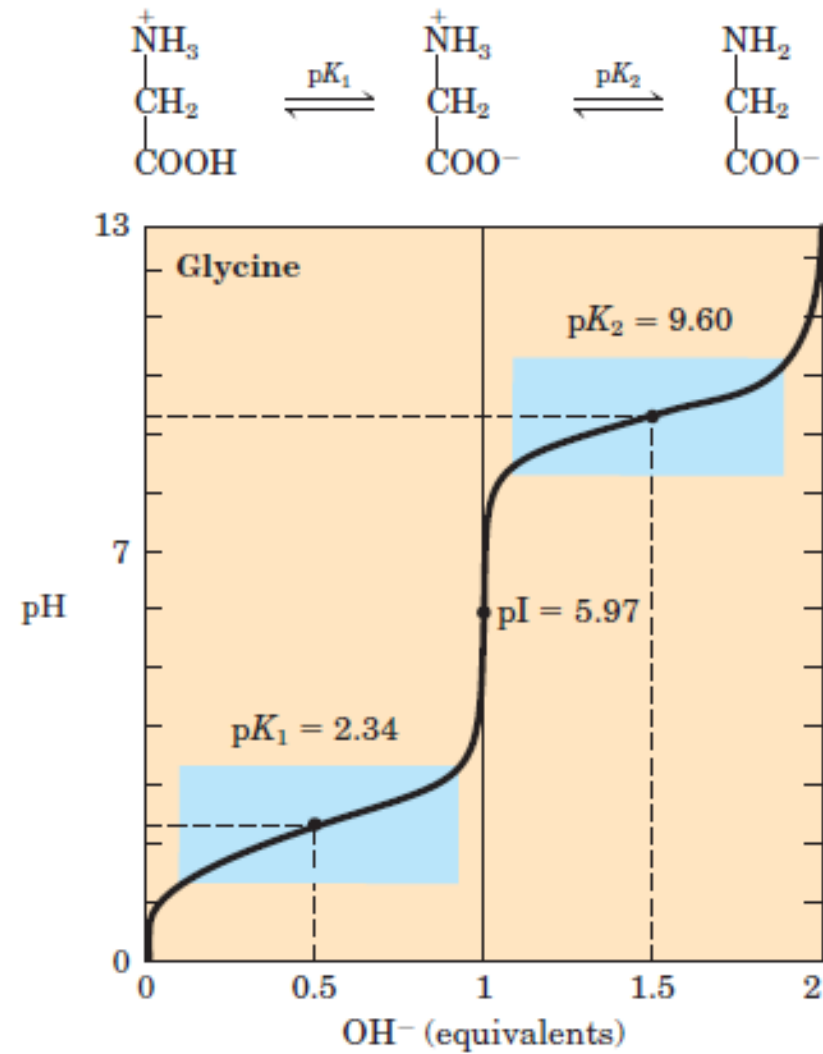
The common amino acids are grouped according to whether their side chains are

acidic
basic
uncharged polar
nonpolar

These 20 amino acids are given both three-letter and one-letter abbreviations.

Thus: alanine = Ala = A

Amino Acids Have Characteristic Titration Curves

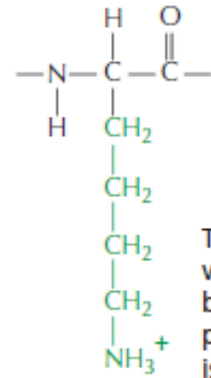


Amino Acids Can Act as Acids and Bases

Amino acids

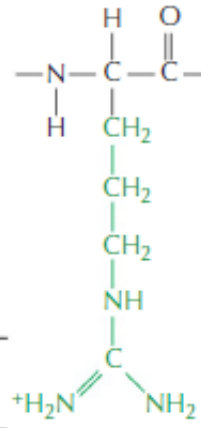
BASIC SIDE CHAINS

lysine
(Lys, or K)

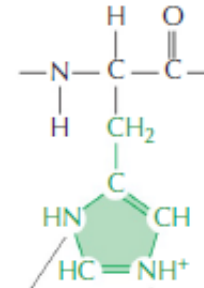


This group is very basic because its positive charge is stabilized by resonance (see Panel 2-1).

arginine
(Arg, or R)



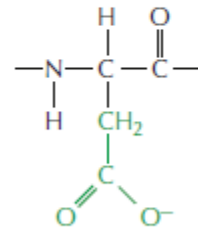
histidine
(His, or H)



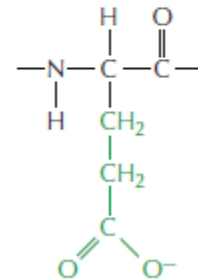
These nitrogens have a relatively weak affinity for an H⁺ and are only partly positive at neutral pH.

ACIDIC SIDE CHAINS

aspartic acid
(Asp, or D)



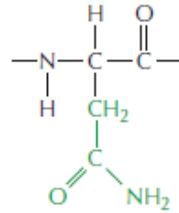
glutamic acid
(Glu, or E)



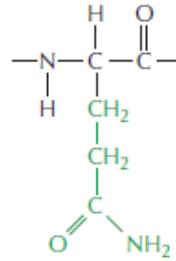
Amino acids

UNCHARGED POLAR SIDE CHAINS

asparagine
(Asn, or N)

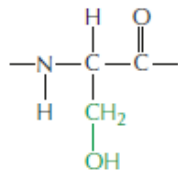


glutamine
(Gln, or Q)

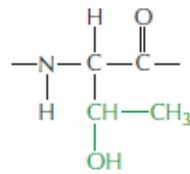


Although the amide N is not charged at neutral pH, it is polar.

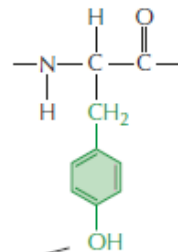
serine
(Ser, or S)



threonine
(Thr, or T)



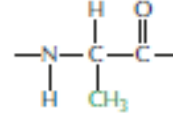
tyrosine
(Tyr, or Y)



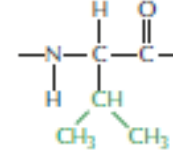
The -OH group is polar.

NONPOLAR SIDE CHAINS

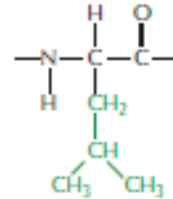
alanine
(Ala, or A)



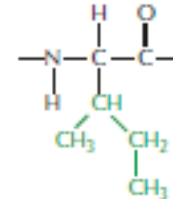
valine
(Val, or V)



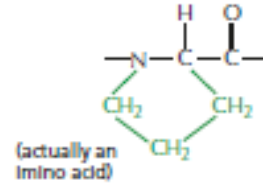
leucine
(Leu, or L)



isoleucine
(Ile, or I)

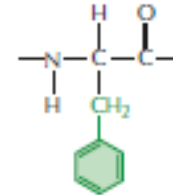


proline
(Pro, or P)

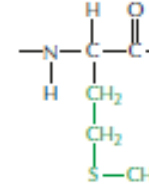


(actually an imino acid)

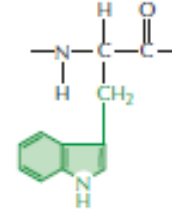
phenylalanine
(Phe, or F)



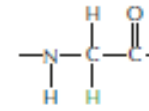
methionine
(Met, or M)



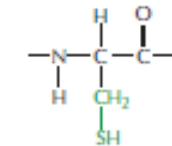
tryptophan
(Trp, or W)



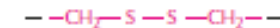
glycine
(Gly, or G)



cysteine
(Cys, or C)



Disulfide bonds can form between two cysteine side chains in proteins.



Amino acids

AMINO ACID		SIDE CHAIN		AMINO ACID		SIDE CHAIN	
Aspartic acid	Asp	D	negative	Alanine	Ala	A	nonpolar
Glutamic acid	Glu	E	negative	Glycine	Gly	G	nonpolar
Arginine	Arg	R	positive	Valine	Val	V	nonpolar
Lysine	Lys	K	positive	Leucine	Leu	L	nonpolar
Histidine	His	H	positive	Isoleucine	Ile	I	nonpolar
Asparagine	Asn	N	uncharged polar	Proline	Pro	P	nonpolar
Glutamine	Gln	Q	uncharged polar	Phenylalanine	Phe	F	nonpolar
Serine	Ser	S	uncharged polar	Methionine	Met	M	nonpolar
Threonine	Thr	T	uncharged polar	Tryptophan	Trp	W	nonpolar
Tyrosine	Tyr	Y	uncharged polar	Cysteine	Cys	C	nonpolar

└──────── POLAR AMINO ACIDS ─────────┐ └──────── NONPOLAR AMINO ACIDS ─────────┐

Six amino acids (CHIMSV), the first letter of the amino acid name is unique

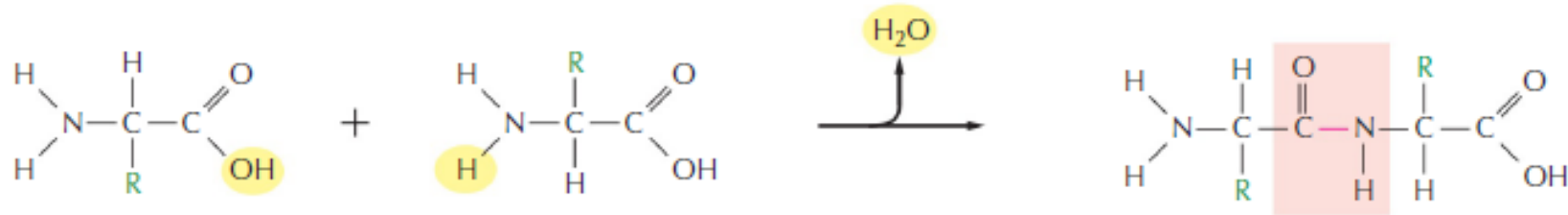
Phonetically suggestive - RFYW: aRginine, Fenylalanine, tYrosine, tWiptophan.

Proteins

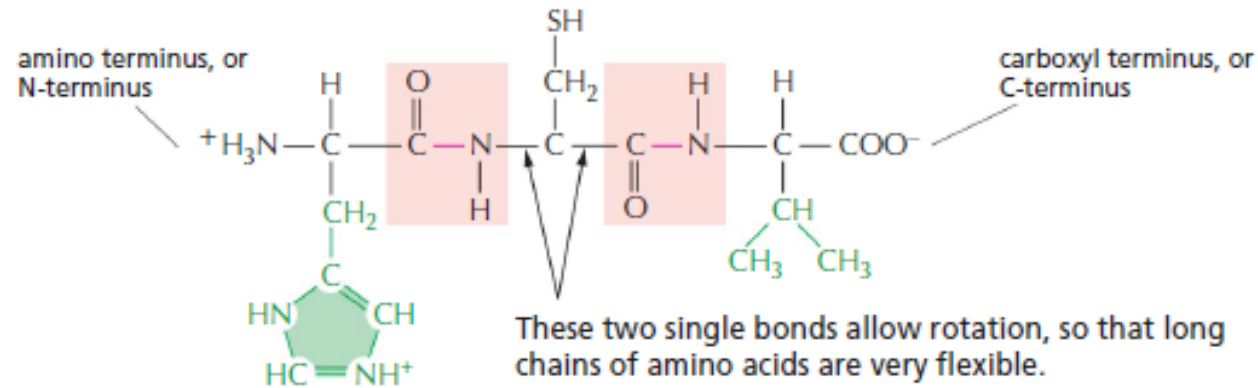
PEPTIDE BONDS

In proteins, amino acids are commonly joined together by an amide linkage, called a peptide bond.

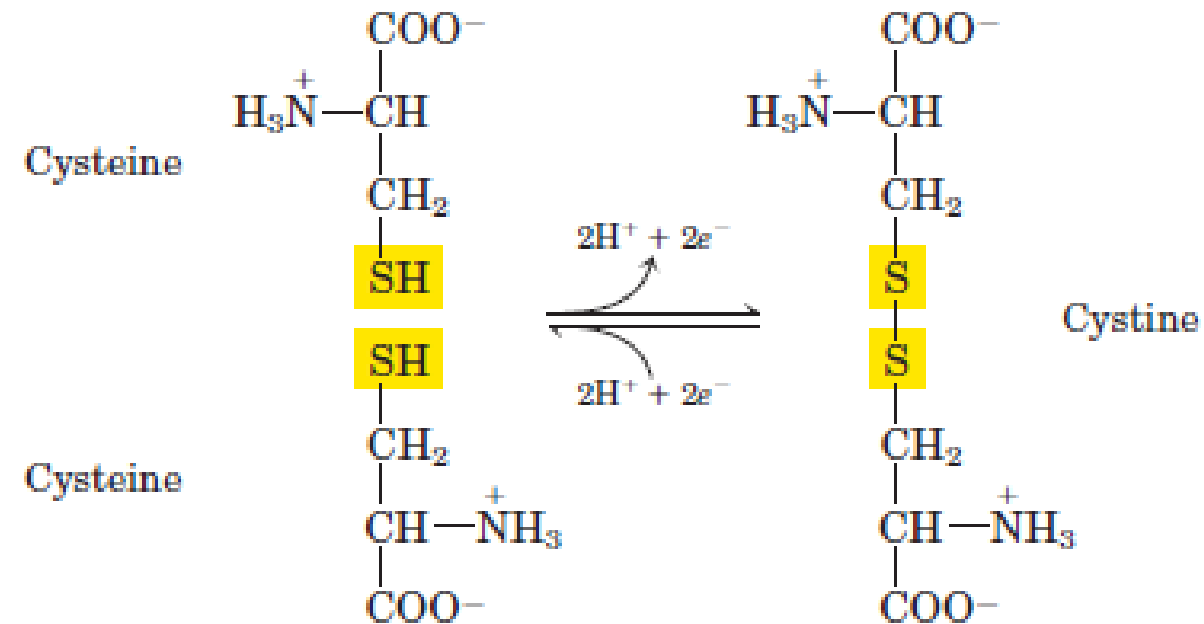
The four atoms in each **peptide bond** (red box) form a rigid planar unit. There is no rotation around the C-N bond.



Proteins are long polymers of amino acids linked by peptide bonds, and they are always written with the N-terminus toward the left. **Peptides** are shorter, usually fewer than 50 amino acids long. The sequence of this tripeptide is histidine-cysteine-valine.



Reversible formation of a disulfide bond by the oxidation of two molecules of cysteine



Molecular Data on Some Proteins

	Molecular weight	Number of residues	Number of polypeptide chains
Cytochrome <i>c</i> (human)	12,400	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (chicken egg white)	14,300	129	1
Myoglobin (equine heart)	16,700	153	1
Chymotrypsin (bovine pancreas)	25,200	241	3
Chymotrypsinogen (bovine)	25,700	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	66,000	609	1
Hexokinase (yeast)	107,900	972	2
RNA polymerase (<i>E. coli</i>)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (<i>E. coli</i>)	619,000	5,628	12
Titin (human)	2,993,000	26,926	1

Some Proteins Contain Chemical Groups Other Than Amino Acids

Class	Prosthetic group	Example
Lipoproteins	Lipids	β_1 -Lipoprotein of blood
Glycoproteins	Carbohydrates	Immunoglobulin G
Phosphoproteins	Phosphate groups	Casein of milk
Hemoproteins	Heme (iron porphyrin)	Hemoglobin
Flavoproteins	Flavin nucleotides	Succinate dehydrogenase
Metalloproteins	Iron	Ferritin
	Zinc	Alcohol dehydrogenase
	Calcium	Calmodulin
	Molybdenum	Dinitrogenase
	Copper	Plastocyanin

Conjugated Proteins